

Homochiral ideas

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Objectives

- Keep the same chirality when building the whole tiling by inflation
- Use only rotations and translations operations: no mirrors !
- (avoid usage of the four key-points in the original inflation rule)
- Keep the monotile shape: $\text{Tile}(a,b) + \text{Tile}(b,a)$
- Use cluster level (full cluster instead of gamma and omega ones ?)
- Make full use of the mapping on a triangular lattice
- Make full use of the ODD and EVEN orientations

Homochiral python code on github

- Under construction !
- <https://github.com/marimperorclerc/homochiral>

python code for drawing clusters with Tile(a,b) in svg graphics.

References

Code is based on the drawsvg package:

<https://pypi.org/project/drawsvg/>

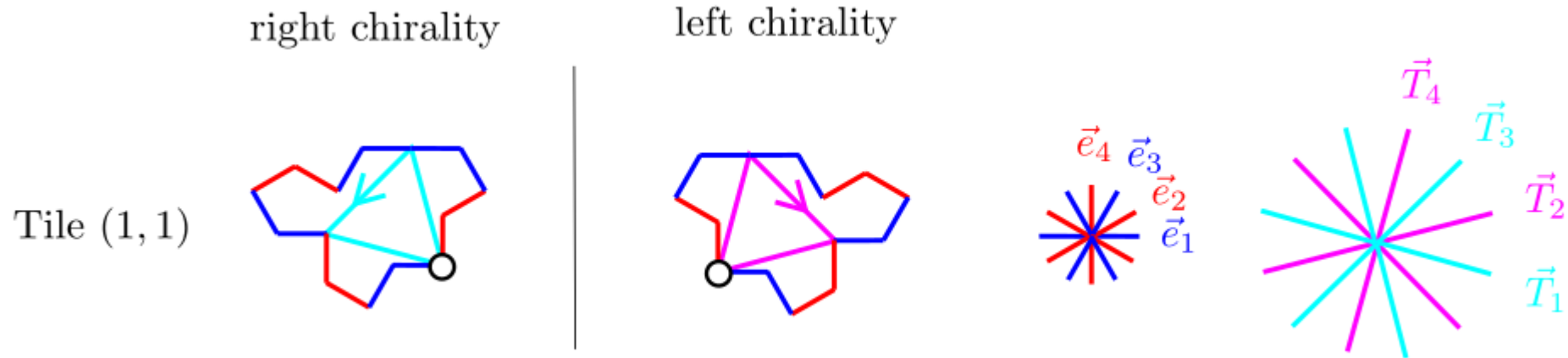
And the python code with alternate chirality for inflation for spectre tiling:

<https://github.com/shrx/spectre/blob/master/spectre.py>

Recent work

- Decomposition using one full cluster with six ODD orientations
- -> find some 'simple' relationships between rotations and translations
- Identify regular triangles
- Link with the trifle+bow-tie decomposition
- $\text{Tile}(a,b) + \text{Tile}(b,a)$ for ODD and EVEN orientations

New reference triangle in Tile(a,b)

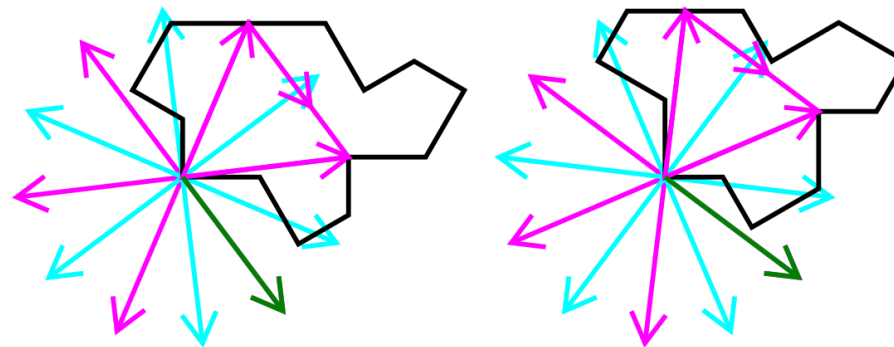


Robust for any T(a,b) tile 😊

Tile(1.6,1.2)

Tile(1.2,1.6)

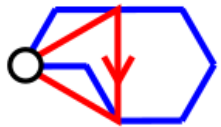
$a=1.6$
 $b=1.2$



New reference triangle in Tile(a,b)

Angle between arrow and x directions:

$\pi/2$

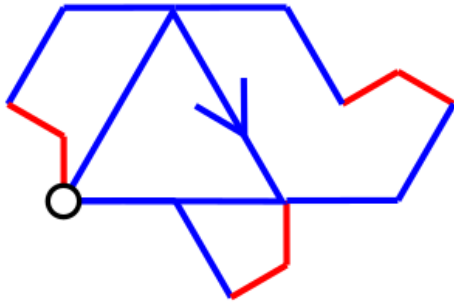


Tile (1, 0)

comet

$\sqrt{3}$

$\pi/3$

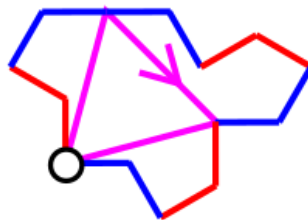


Tile ($\sqrt{3}$, 1)

turtle

$2\sqrt{3}$

$\pi/4$

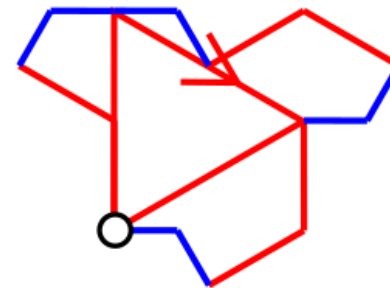


Tile (1, 1)

spectre

$\sqrt{6}$

$\pi/6$



Tile (1, $\sqrt{3}$)

hat

$2\sqrt{3}$

0



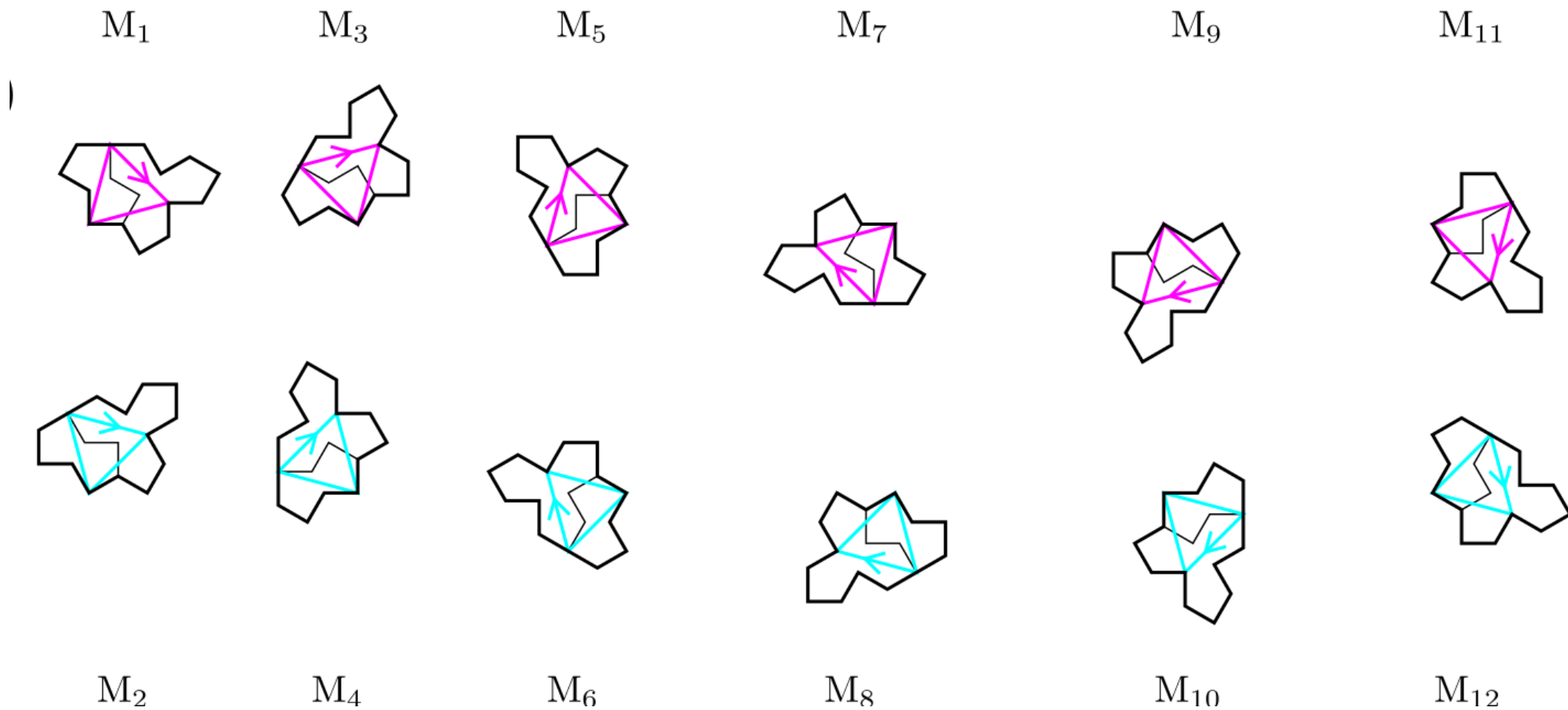
Tile (0, 1)

chevron

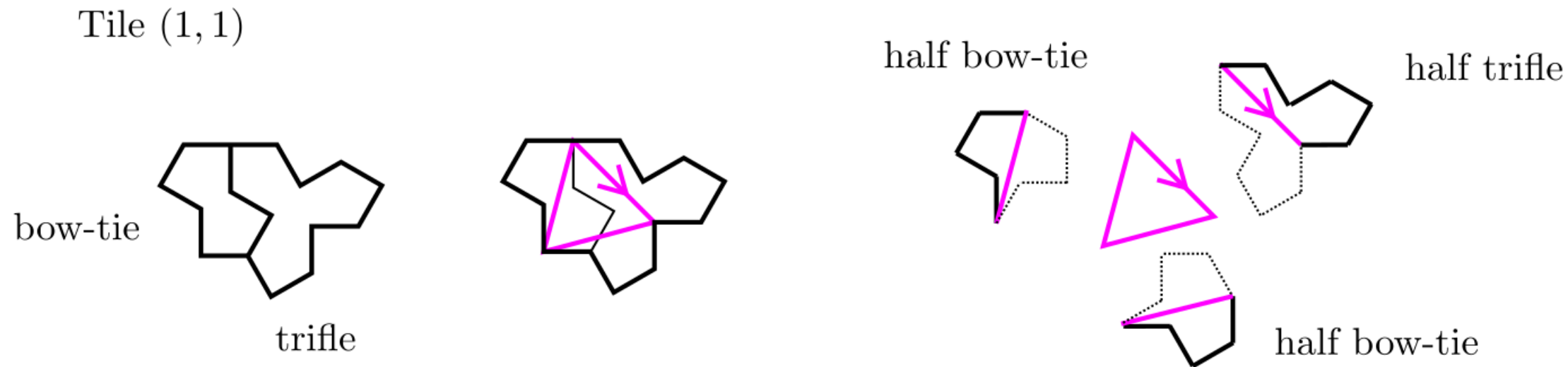
$\sqrt{3}$

Edge length of the regular triangle = $\sqrt{3} \cdot \sqrt{a^2 + b^2}$

ODD and EVEN orientations for Tile(1,1)



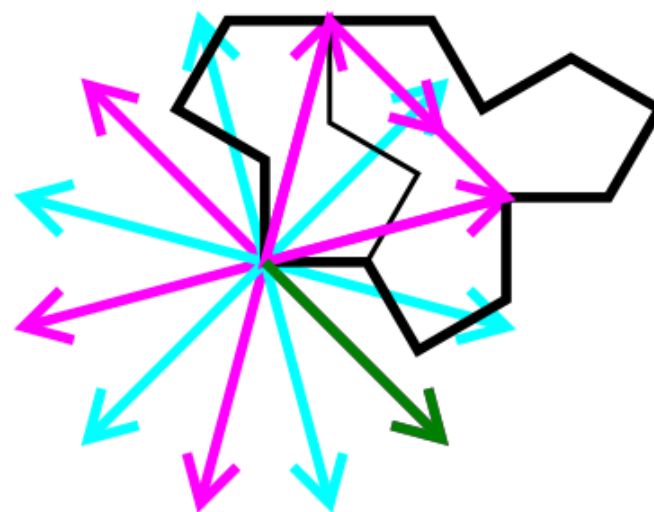
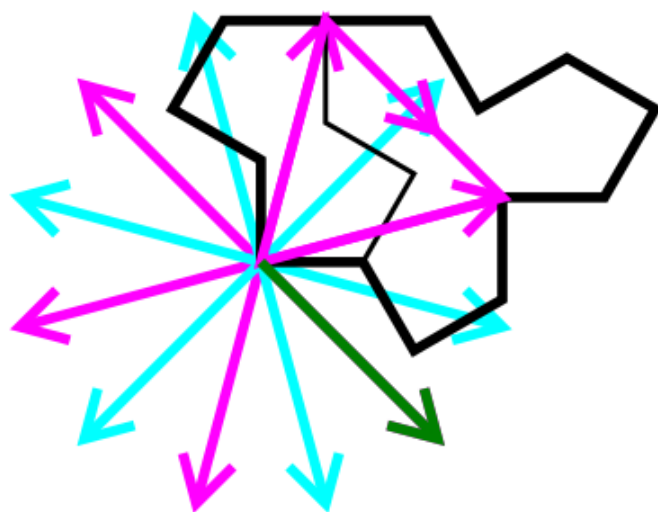
Link with trifle and bow-tie in $\text{Tile}(1,1)$



Tile(1,1) : spectre tile

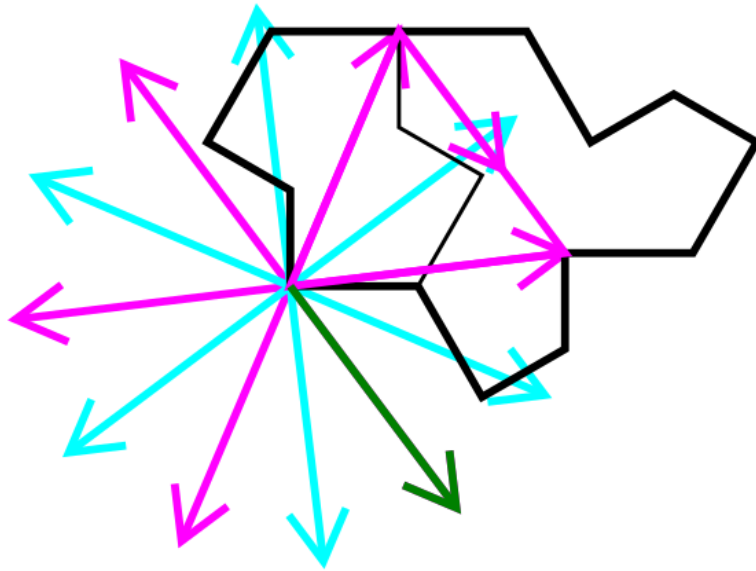
Tile(1.0,1.0)

Tile(1.0,1.0)

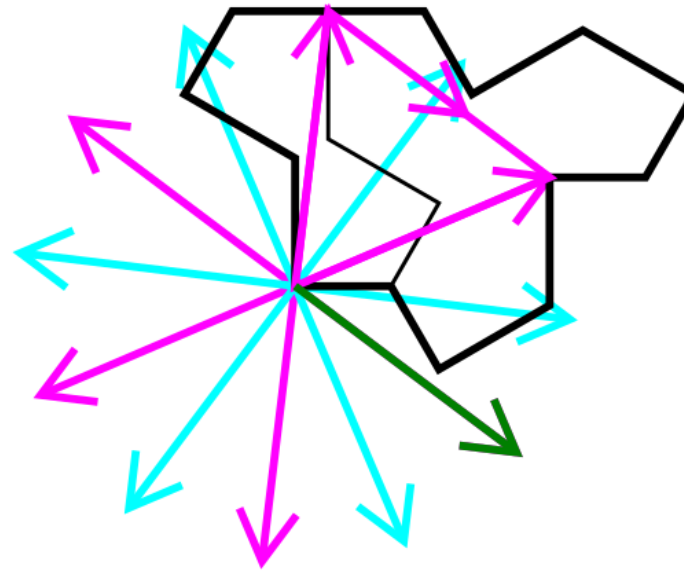


Tile(a,b) and Tile(b,a)

Tile(1.6,1.2)



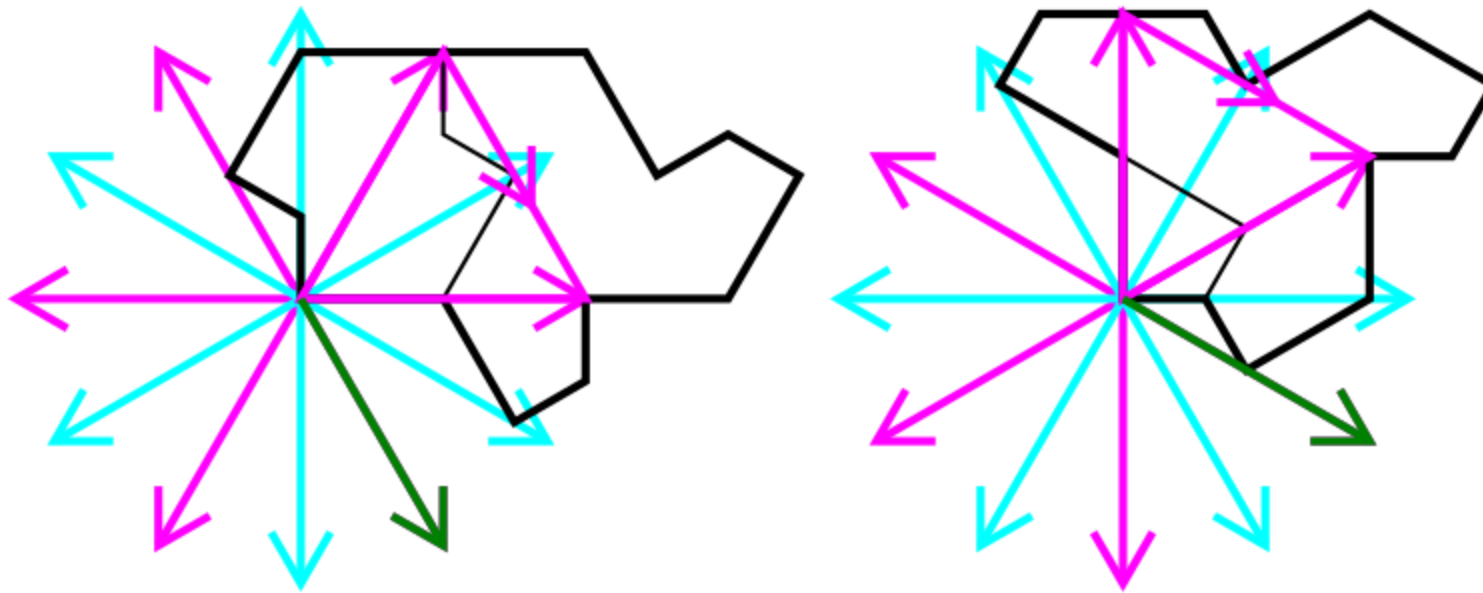
Tile(1.2,1.6)



- Thin and fat bow-ties
- Trifle gives left and right chiral shapes
- The 12 base vectors deduce from a vertical mirror reflection

Tile(a,b) and Tile(b,a) for turtle and hat

Tile(1.73,1.0) Tile(1.0,1.73)

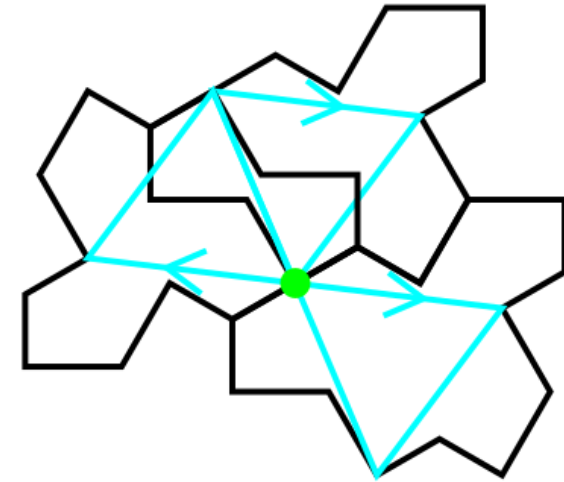
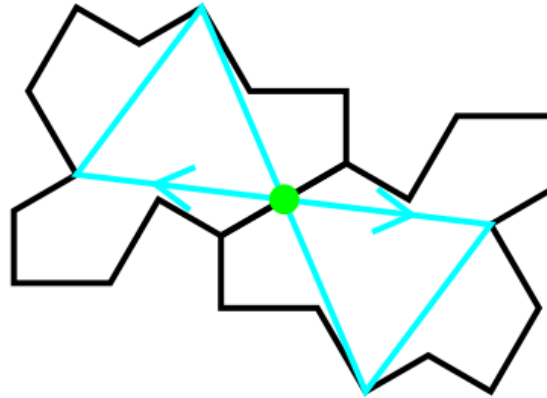
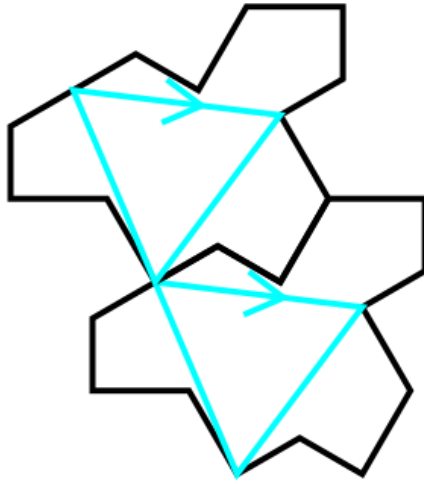


- Trifle gives left and right chiral shapes
- The 12 base vectors deduce from a vertical mirror reflection

Duplicated versus inverted for any Tile(a,b)

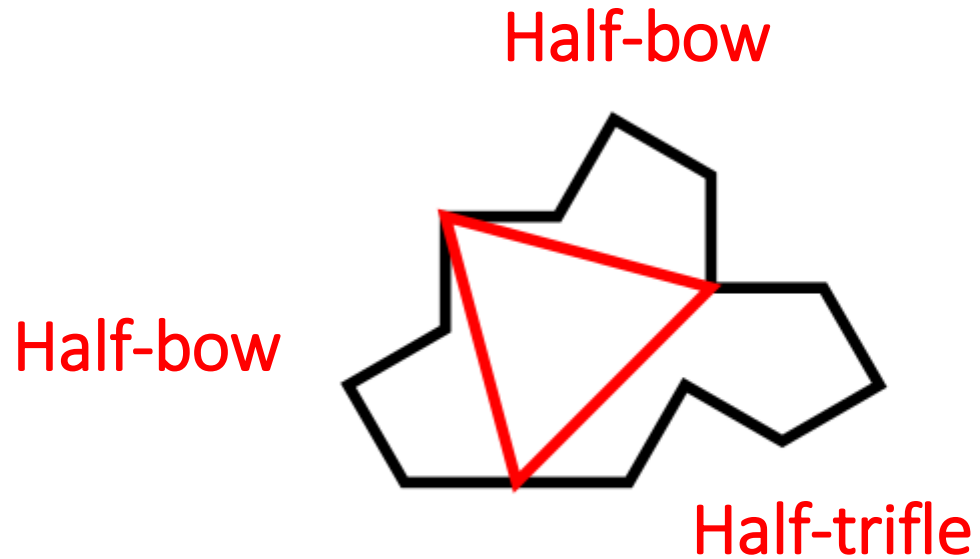
$a=1.6$

$b=1.2$

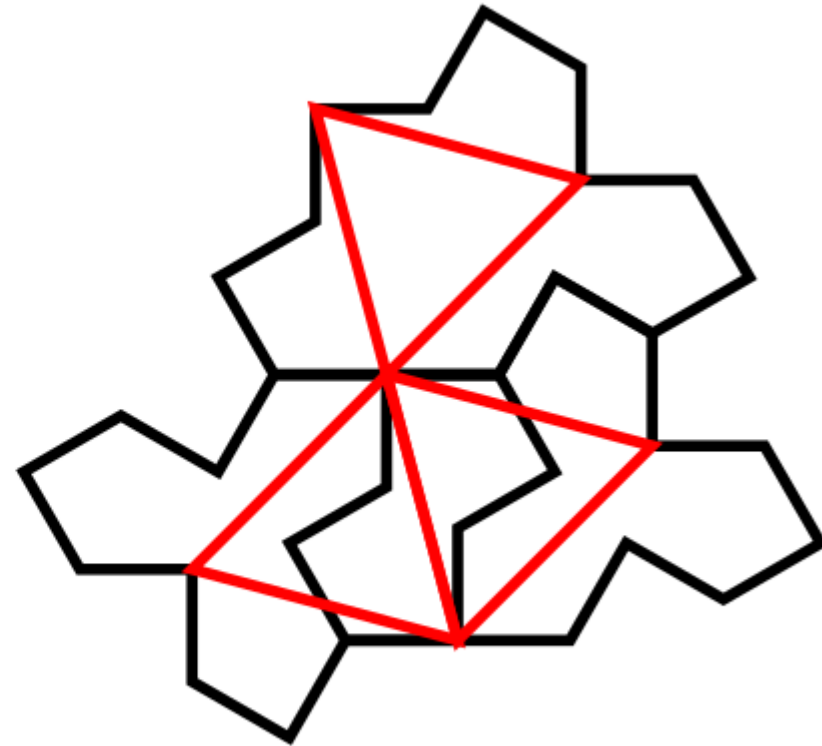


Duplicated tile versus inverted tile

Duplicated versus inverted for right Tile(1,1)



2 other shapes are left:
-> 2 half-bows
-> 1 half-trifle

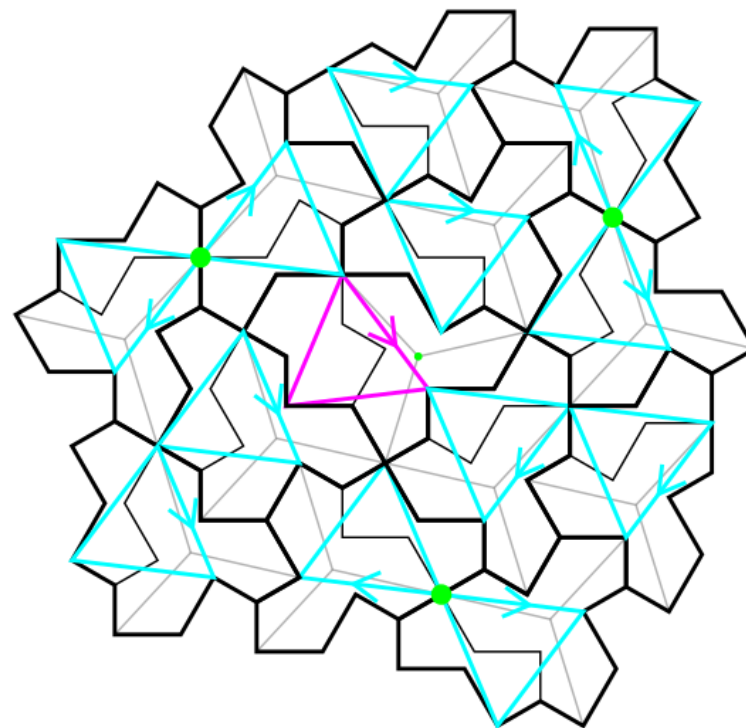
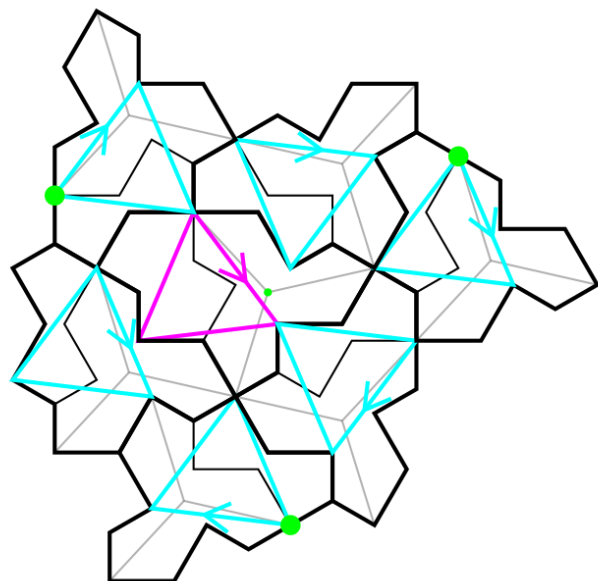


Duplicated tile versus
inverted tile

Cluster and full cluster

$a=1.6$

$b=1.2$

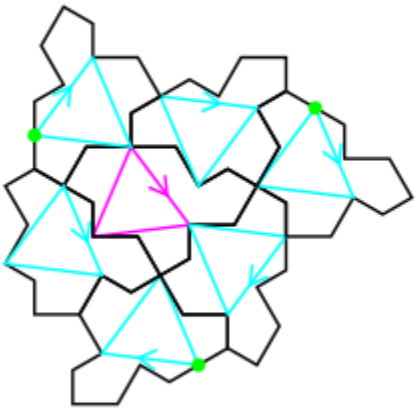


Green points always form a regular triangle 😊

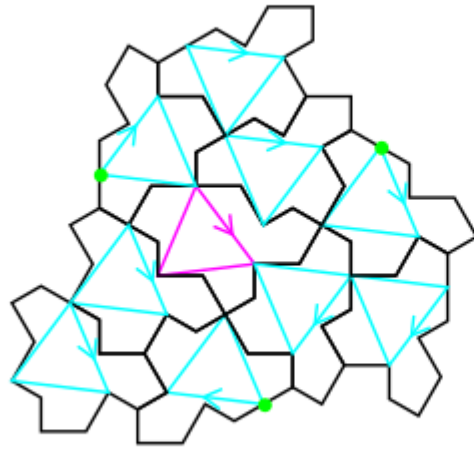
Cluster and full cluster

$$a=1.6$$

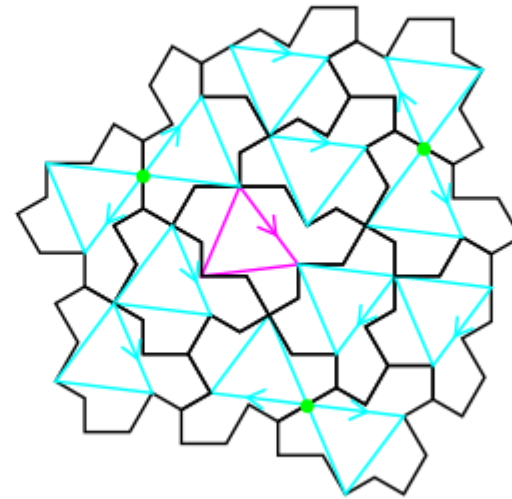
$$b=1.2$$



C cluster
(corona around a
ODD tile)



The three
duplicated tiles
are all in the tiling



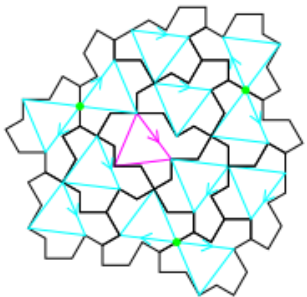
Three more possible tiles
using inversion around the
green points

full cluster with its six rotations: $n(2\pi/6)$

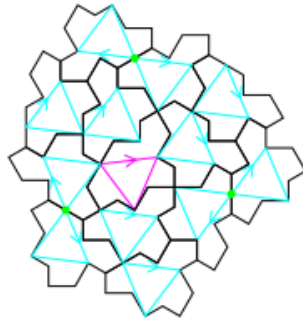
$a=1.6$

$b=1.2$

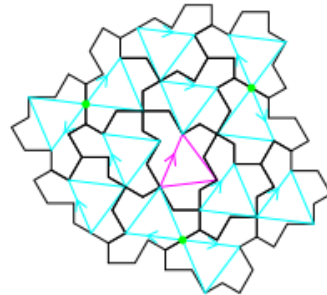
C1



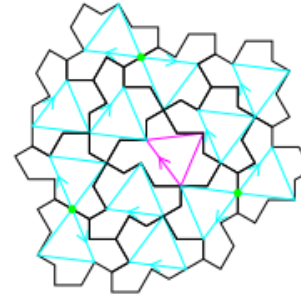
C2



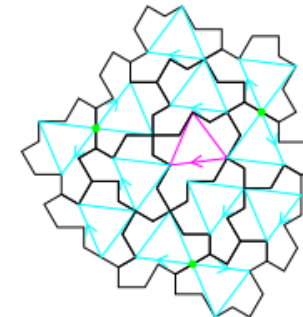
C3



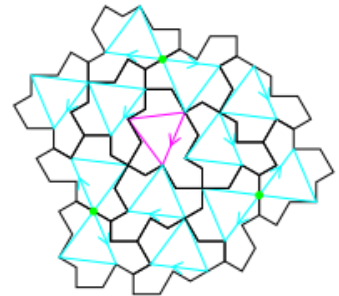
C4



C5



C6



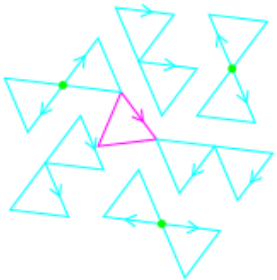
- TO DO: code assembly rules using translations along triangle edges !
- Link between translations and relative ODD orientations ???
- -> combinatory properties

full cluster with its six rotations: $n(2\pi/6)$

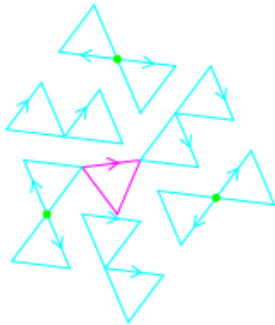
$a=1.6$

$b=1.2$

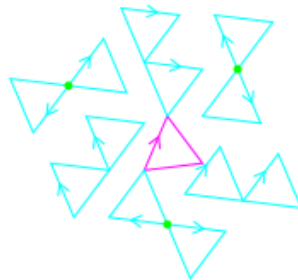
C1



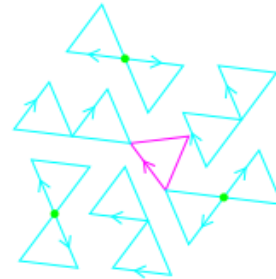
C2



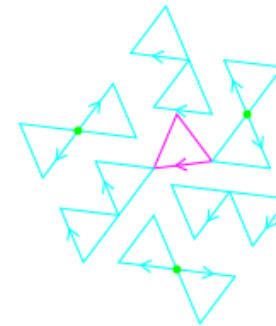
C3



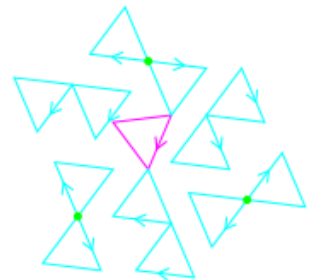
C4



C5

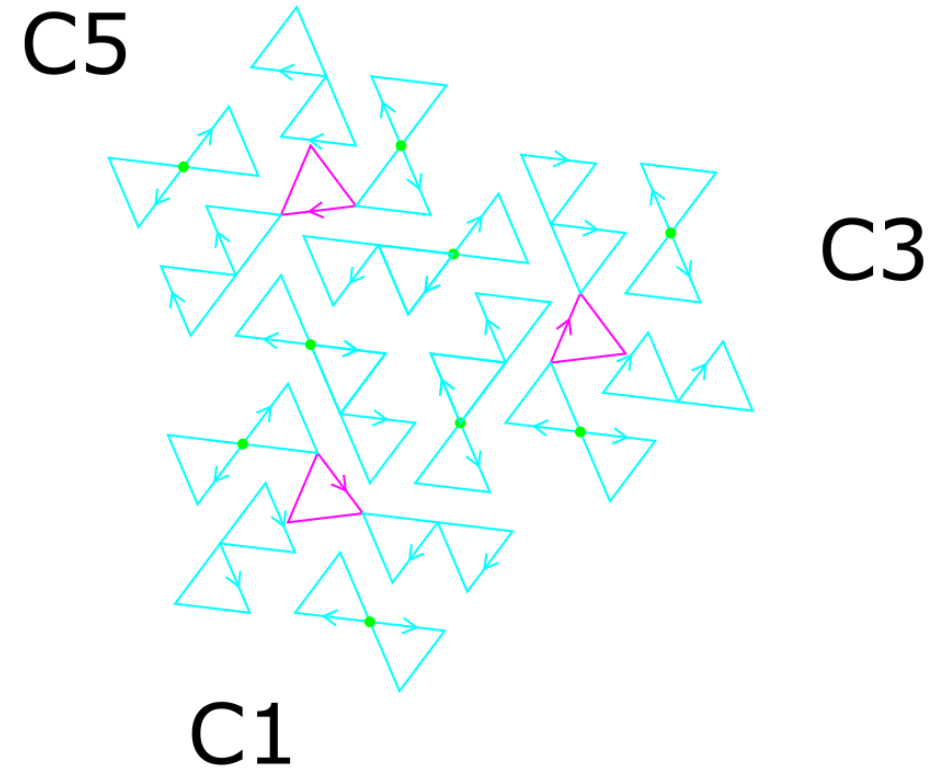
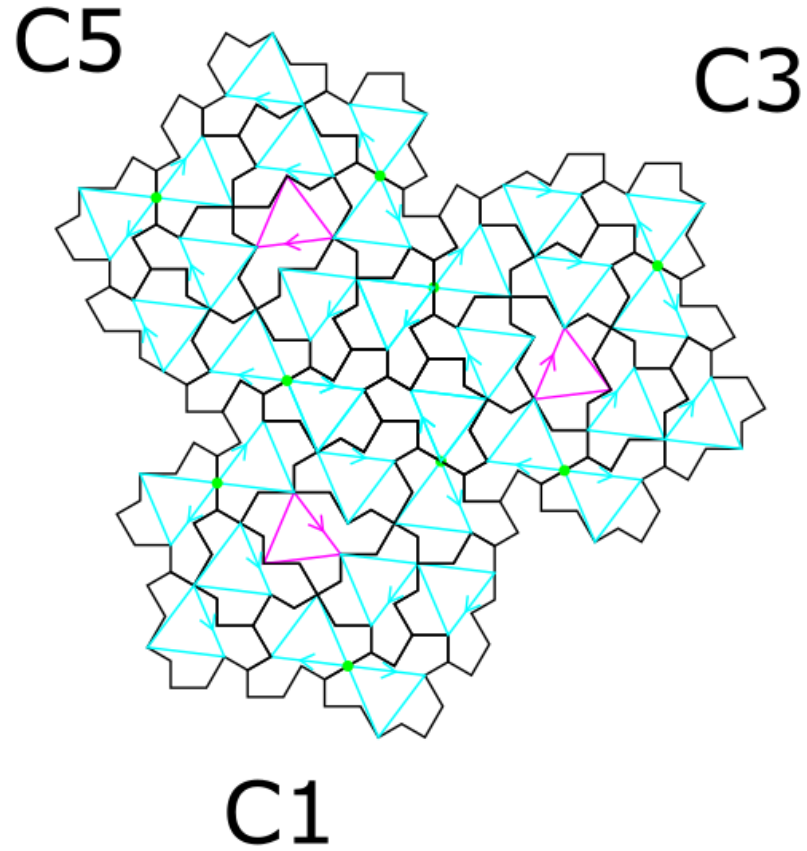


C6

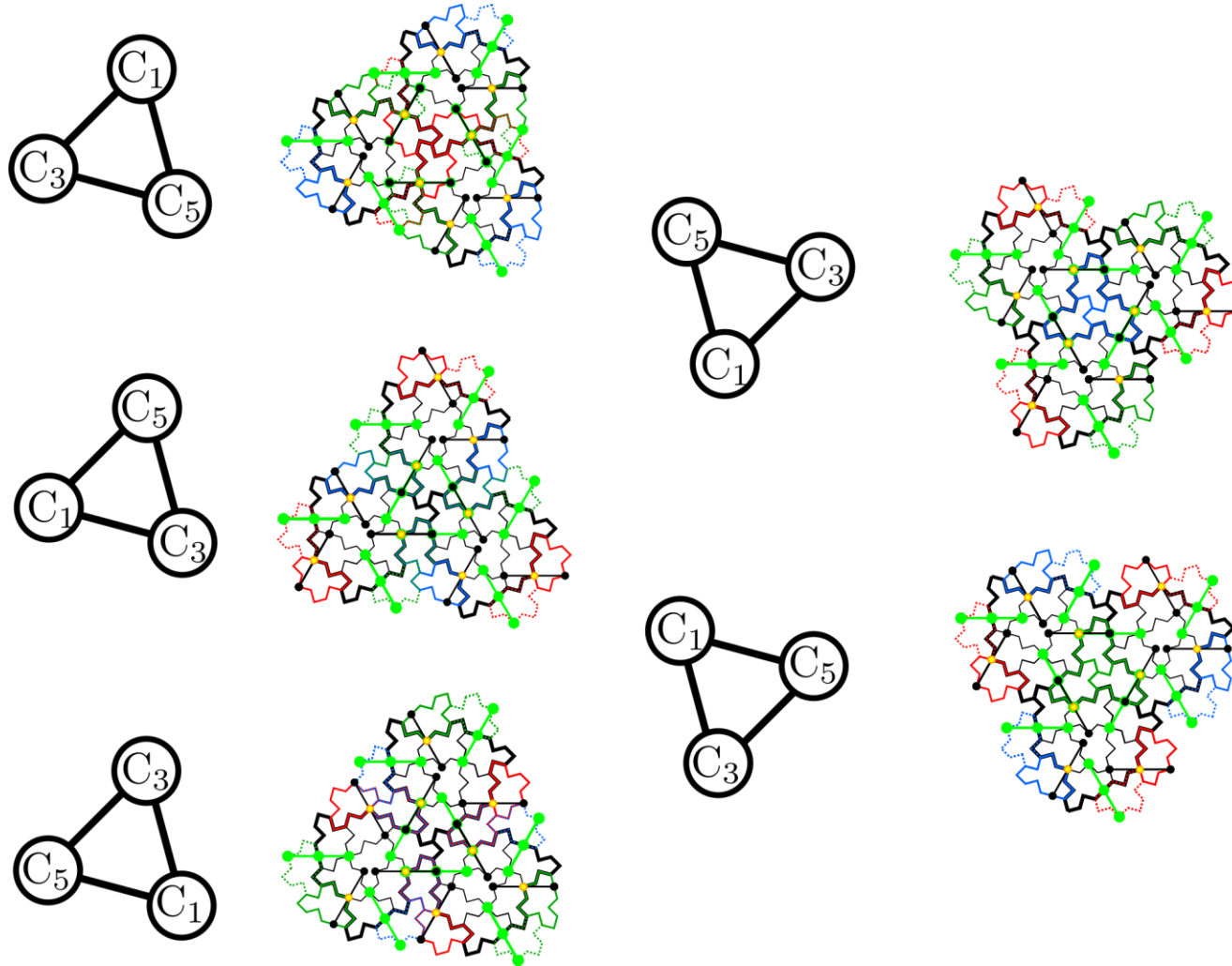


- TO DO: code assembly rules using translations along triangle edges !
- Link between translations and relative ODD orientations ???
- -> combinatory properties

Some combination of three clusters

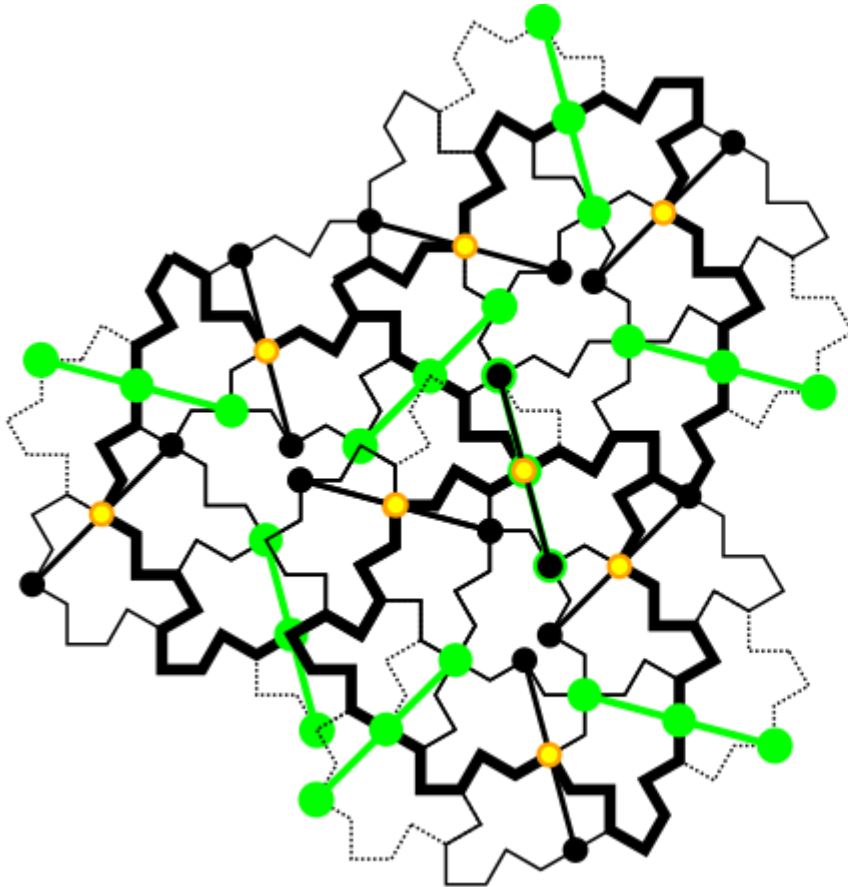


C1, C3 ,C5 triangle : 5 possible combinations

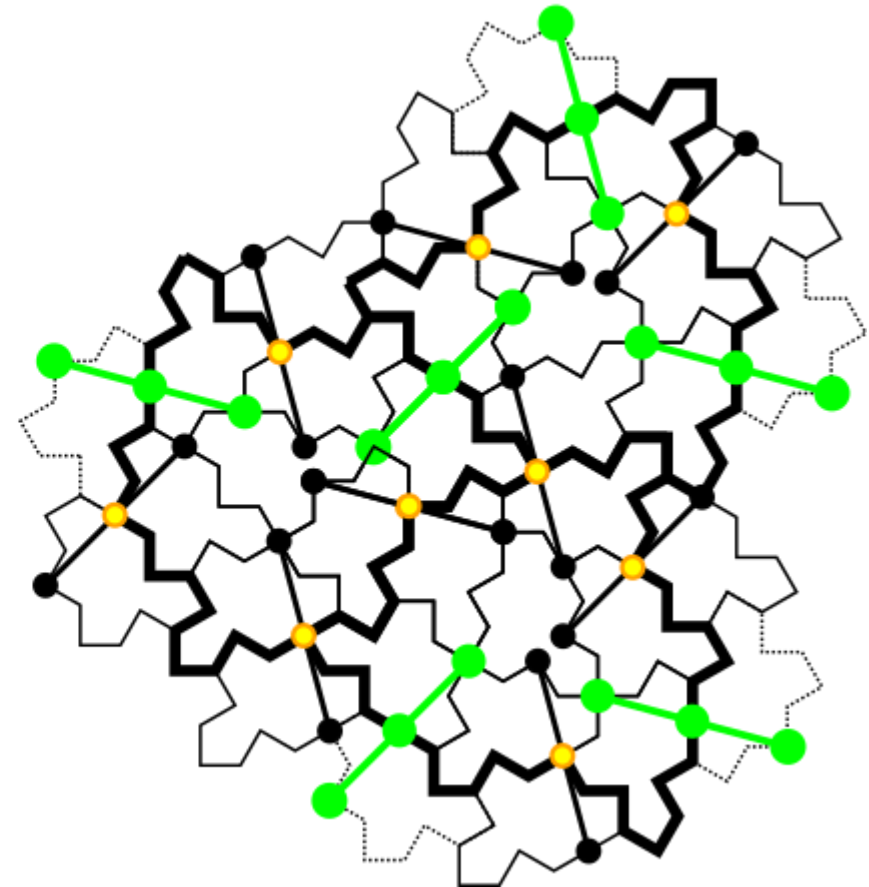


- Possible seed for homochiral inflation ???

Defect healing by keeping the black line when superposition with a green one

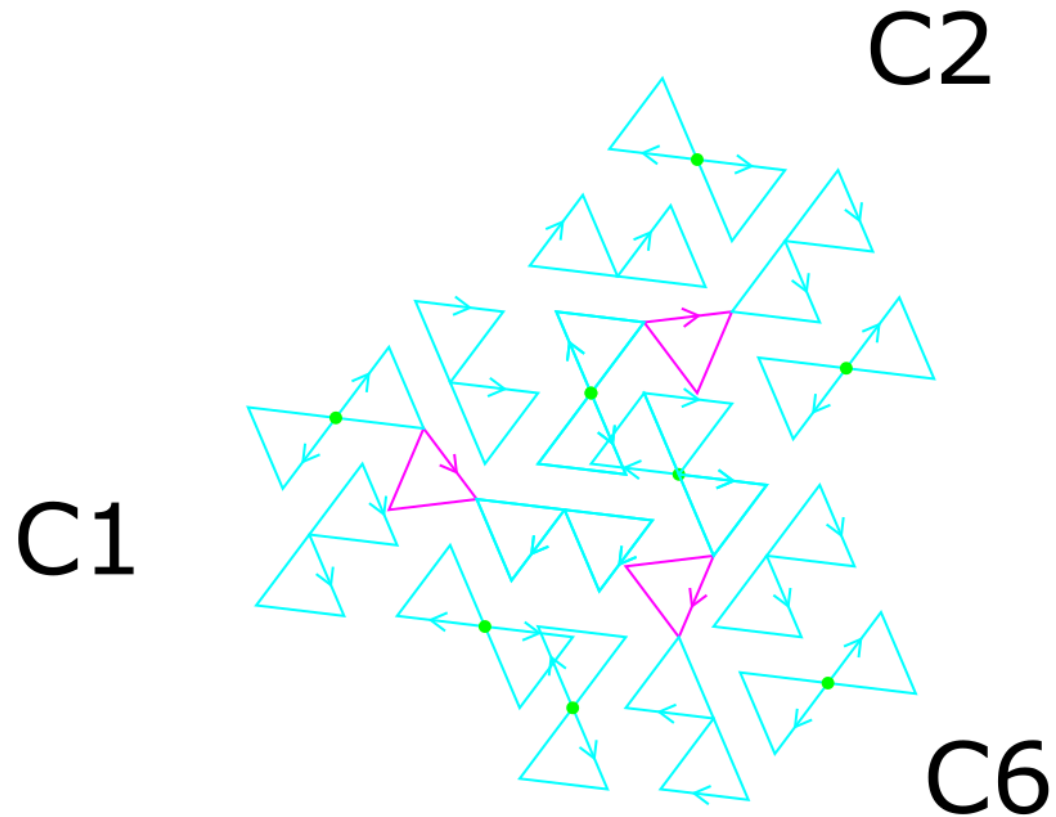


- Starting point for worms 😊

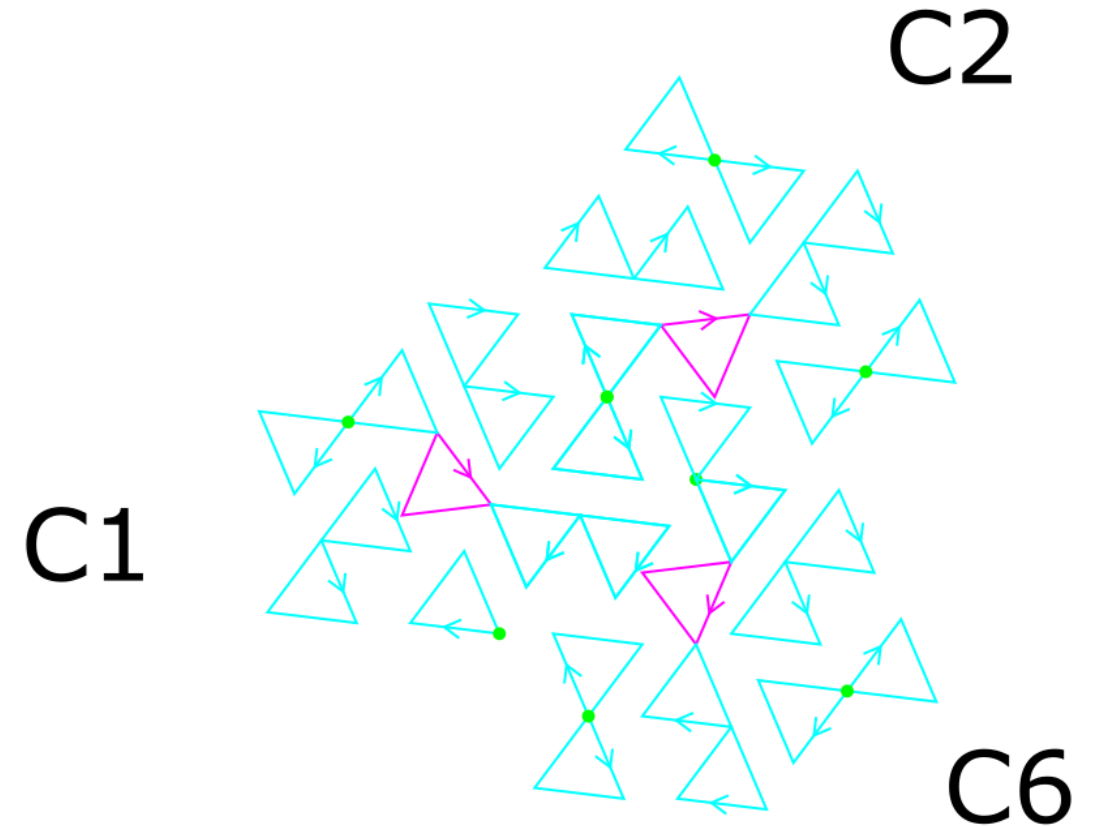


- After removing two inverted tiles

Defect healing

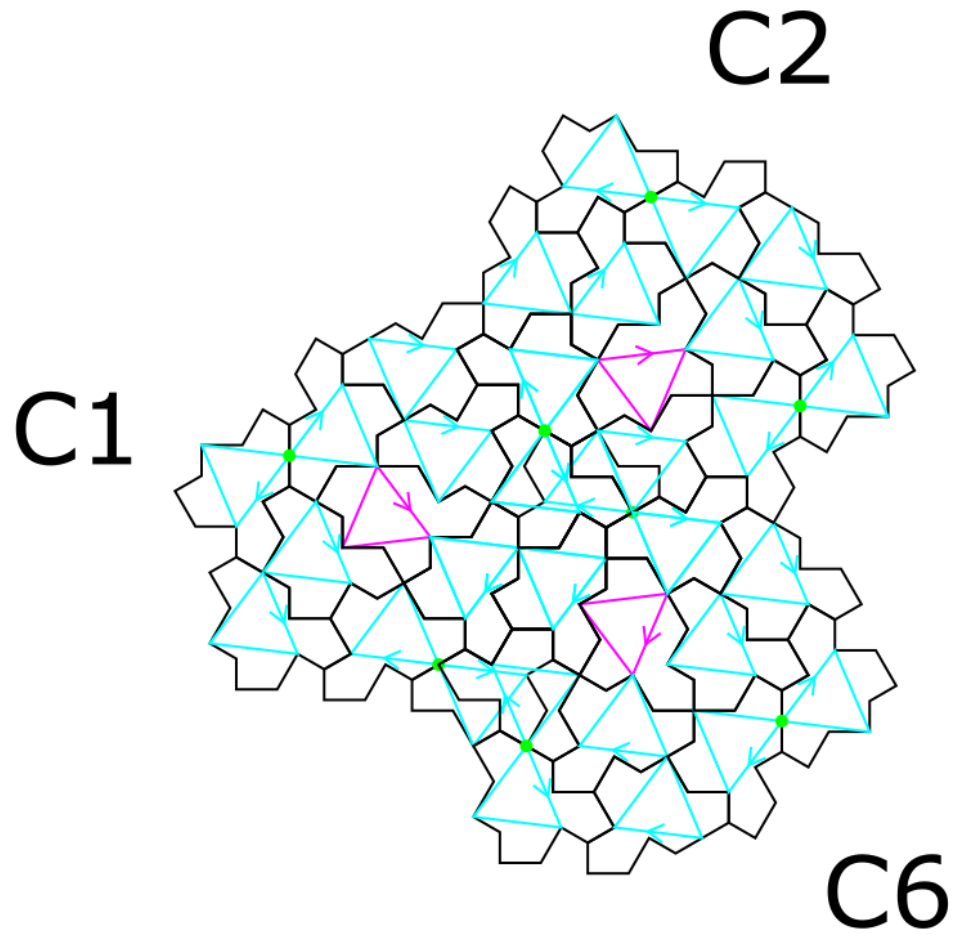


- Starting point for worms 😊

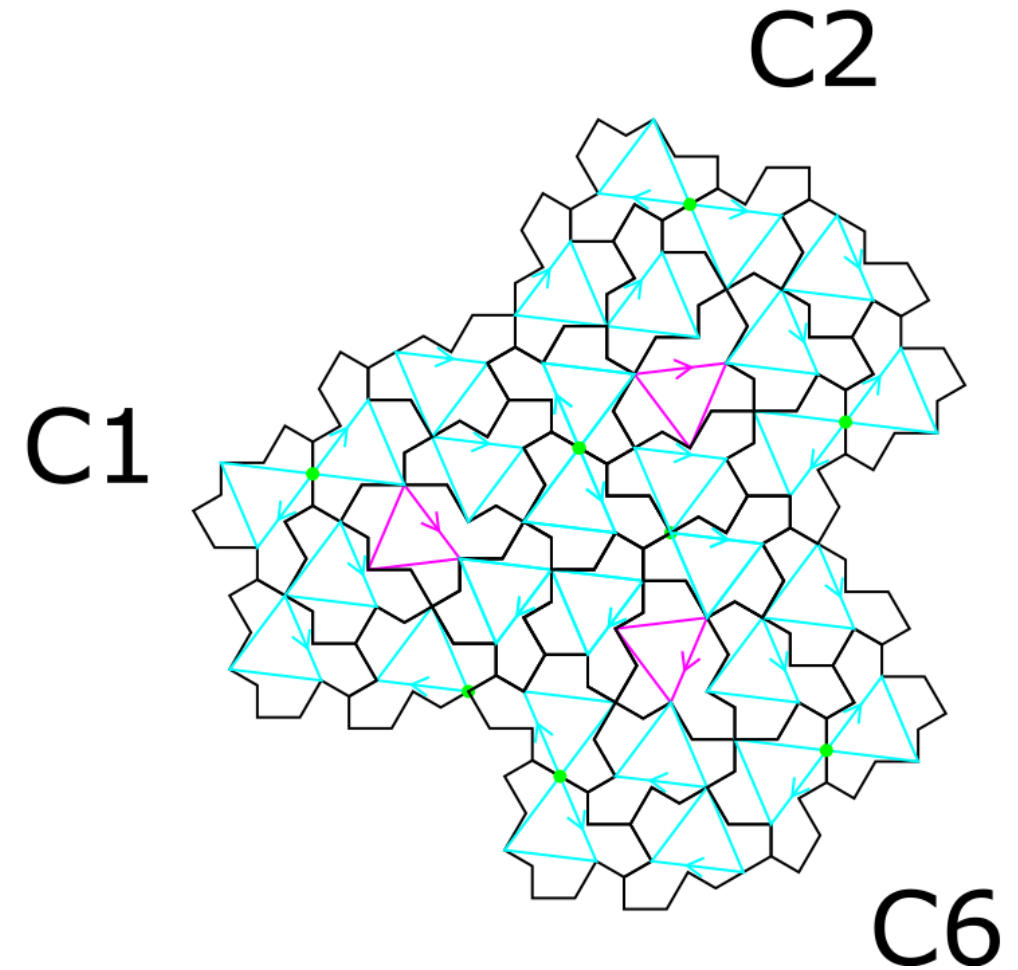


- After removing two inverted tiles

Defect healing

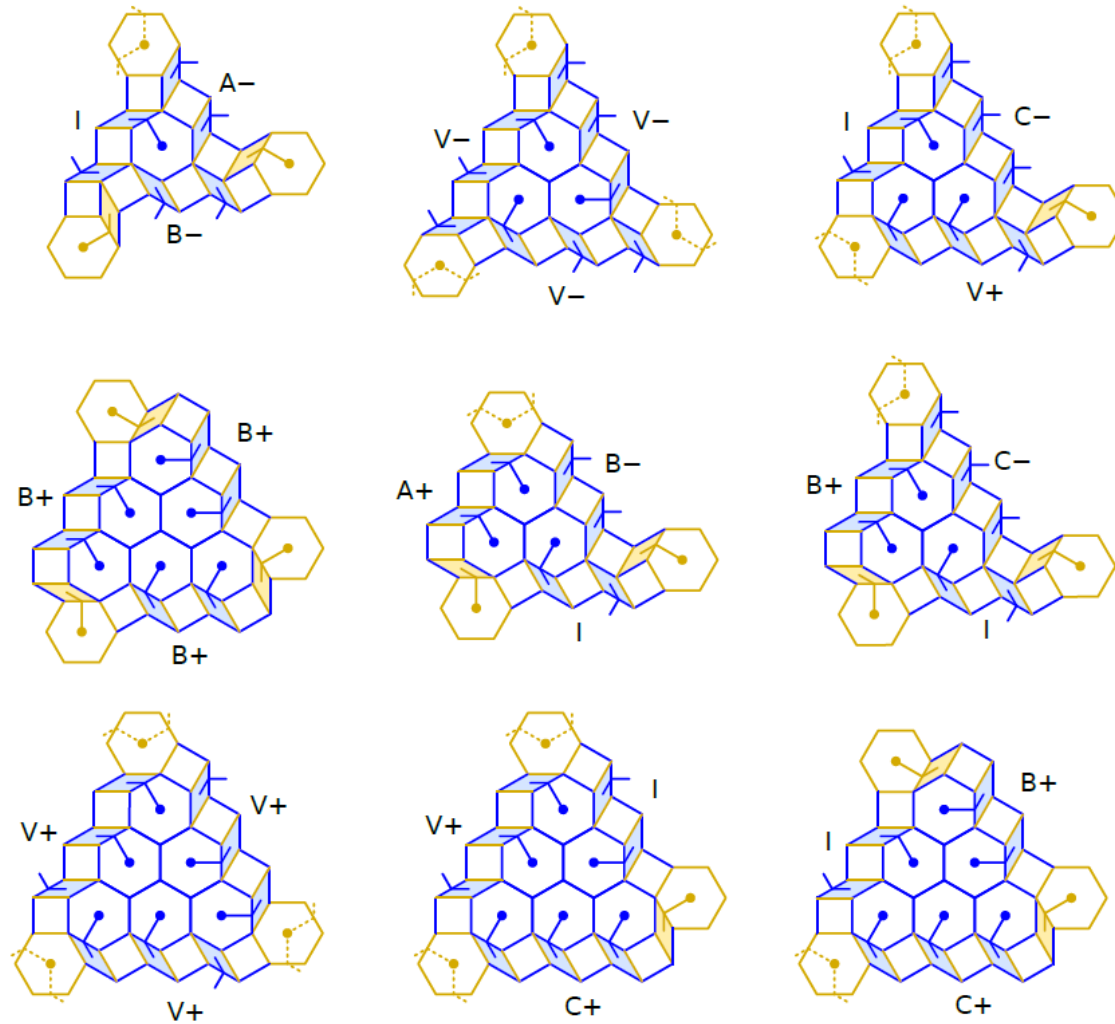


- Starting point for worms ☺



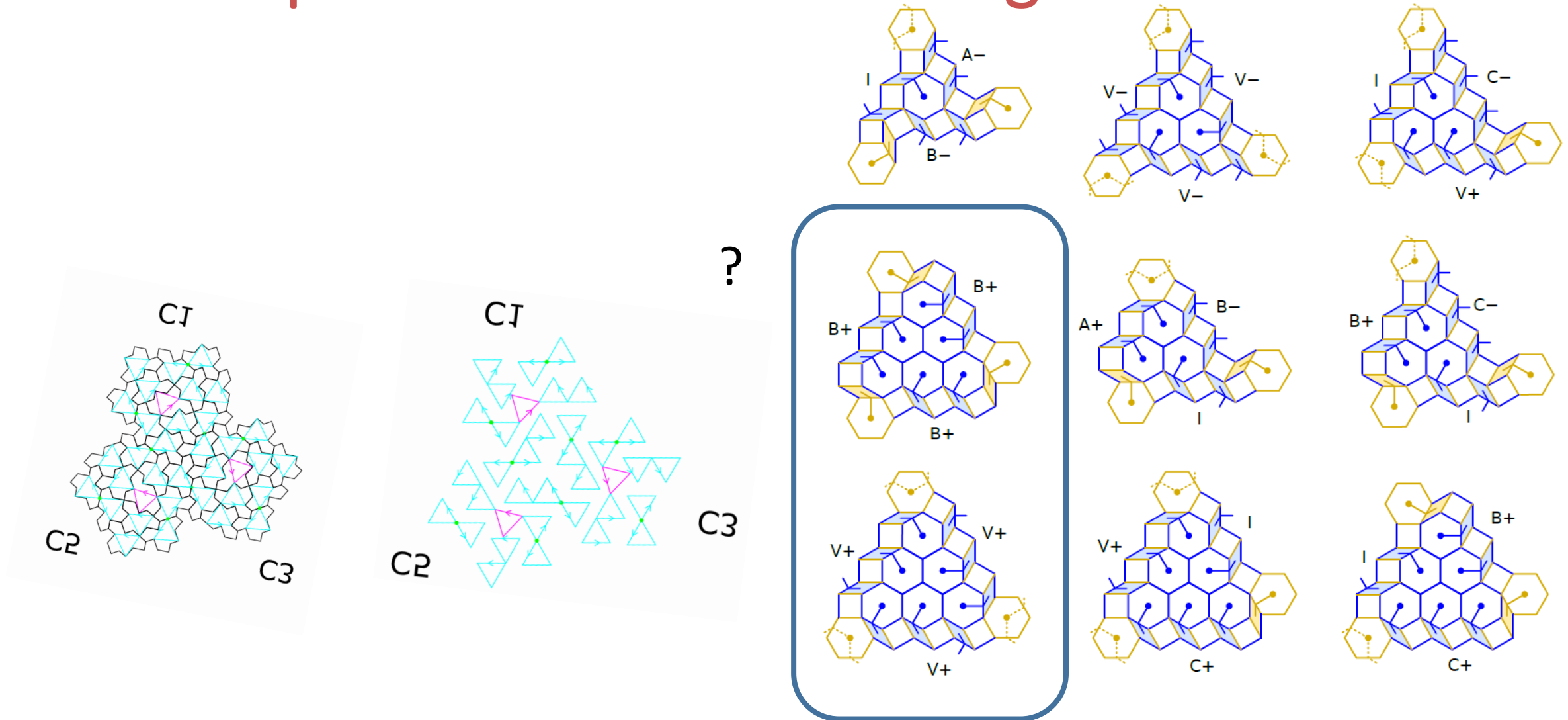
- After removing two inverted tiles

Types of hexagonal clusters



Combinatorial properties

Find equivalence with the hexagonal clusters ?

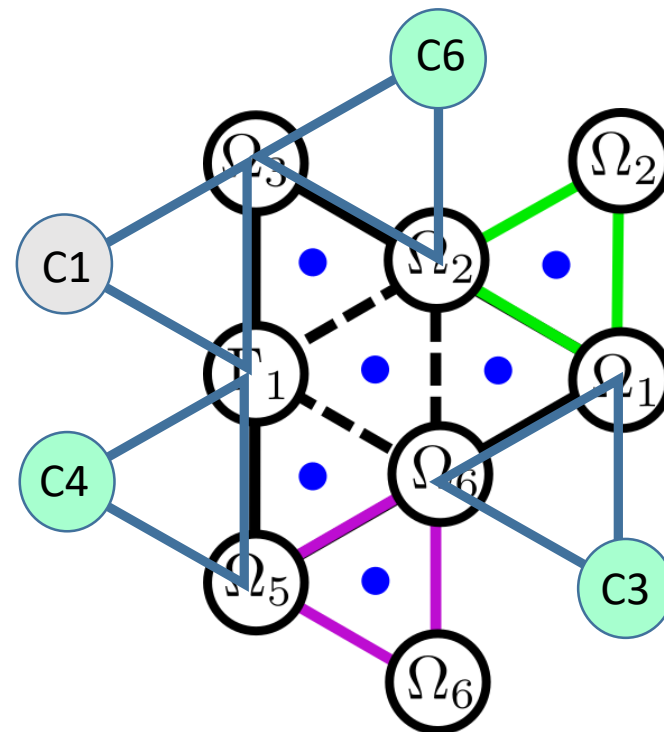
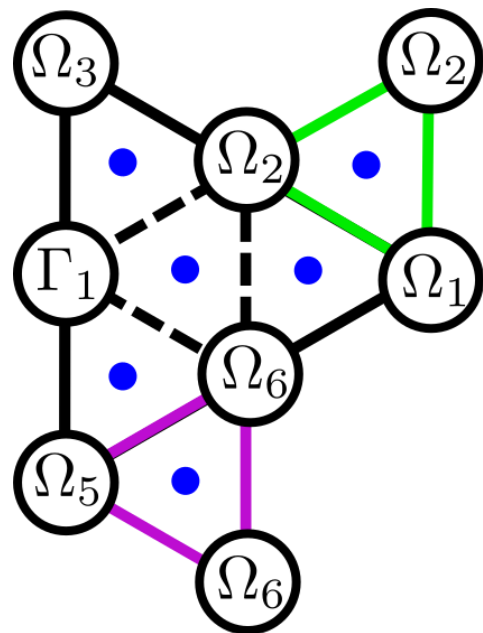


Chéritat, A., (2024). Observations on the hex clusters of the spectre tilings. <https://arxiv.org/abs/2407.05359>

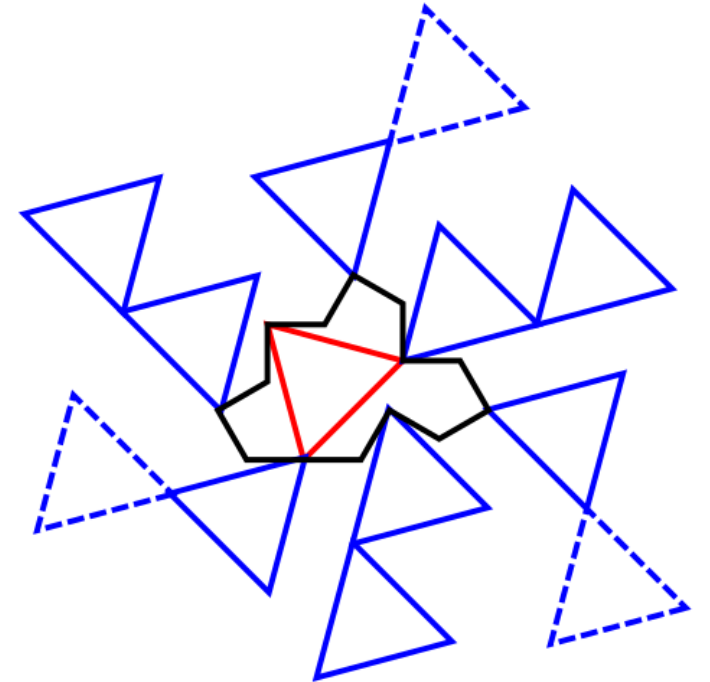
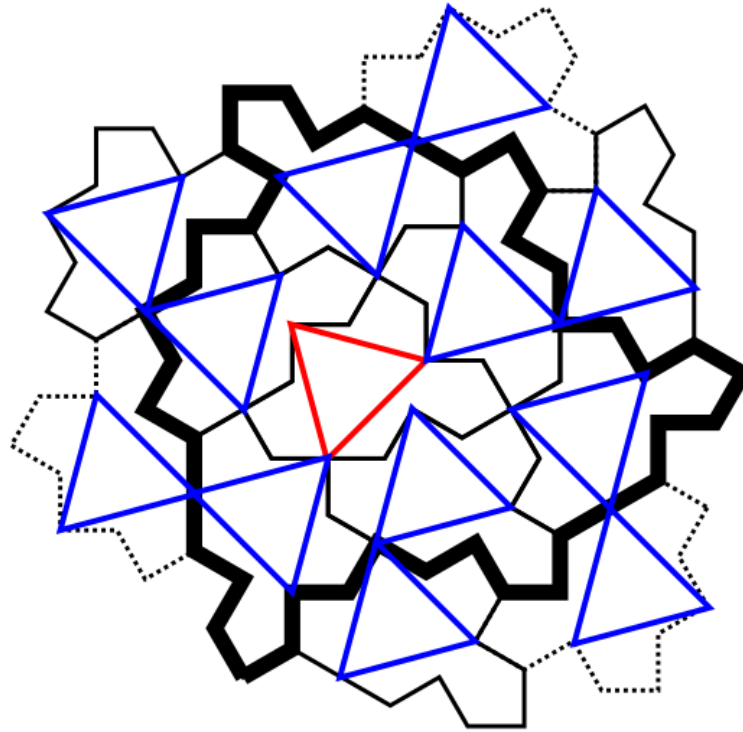
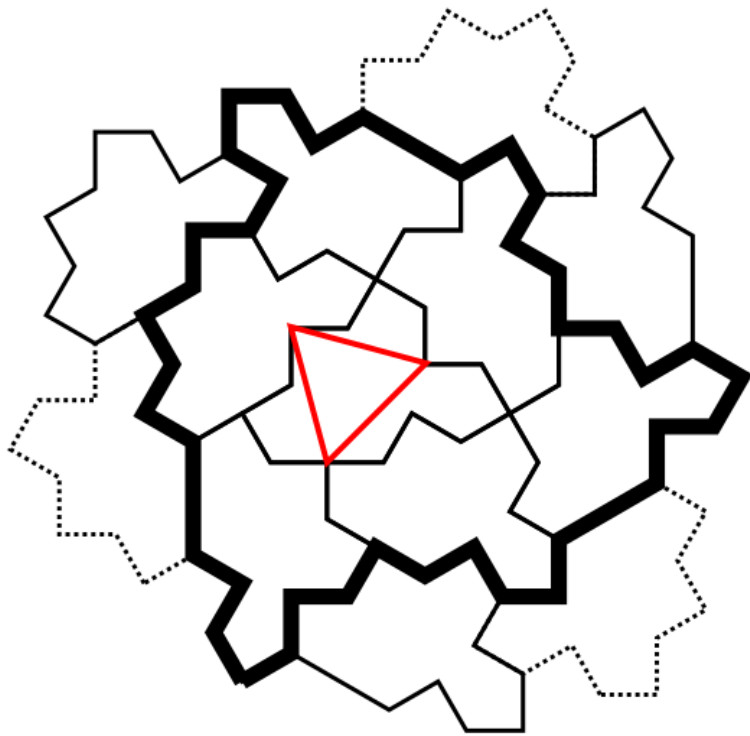
Further ideas ...

- Use cluster level
- For rotation part: CODE the sequence of the six rotations in the C cluster to be repeated at larger length scales
- For translation part: use the duplicated tiles along the reference triangles and the inverted tiles around the three green points
- Hypothesis: rule is that when two duplicated tiles superimposed with two inverted tiles at a green point, one inverted tile is omitted
- Use the green points to assemble the clusters ???

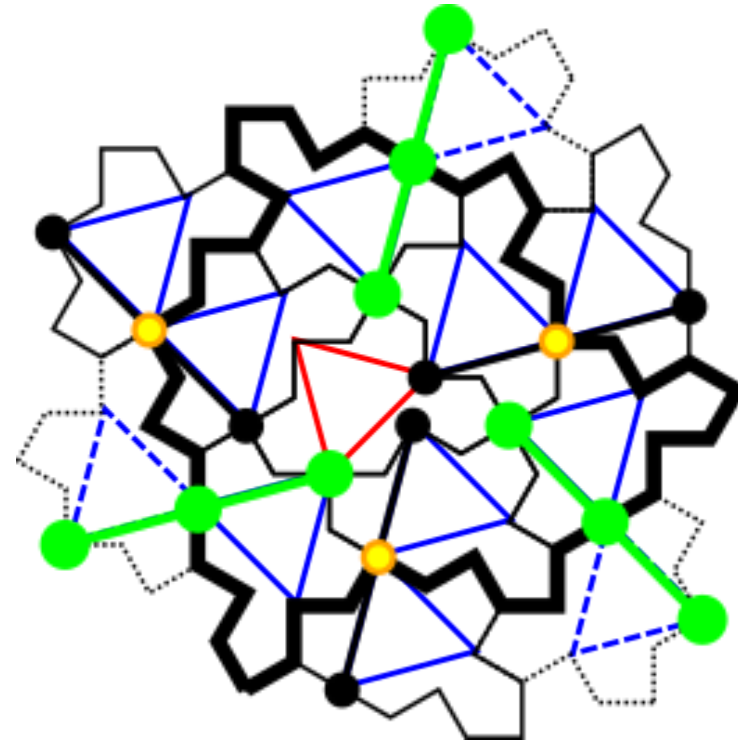
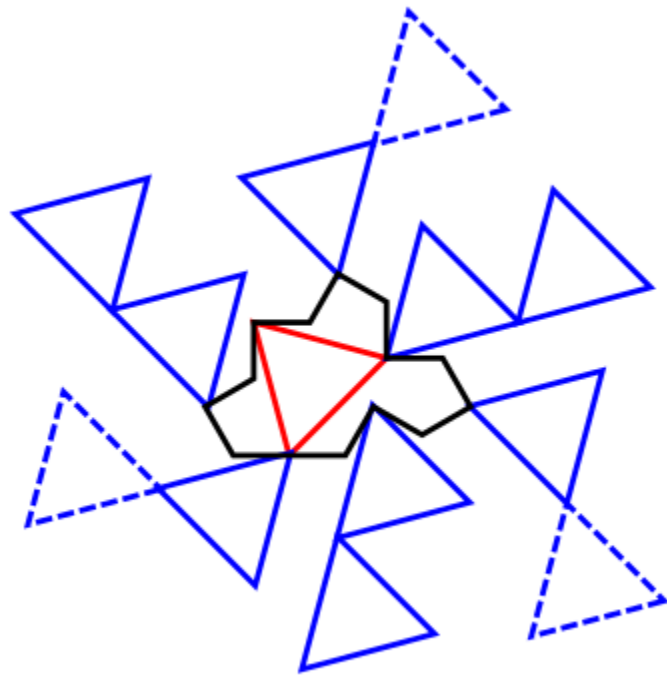
Add the full cluster at this level ???



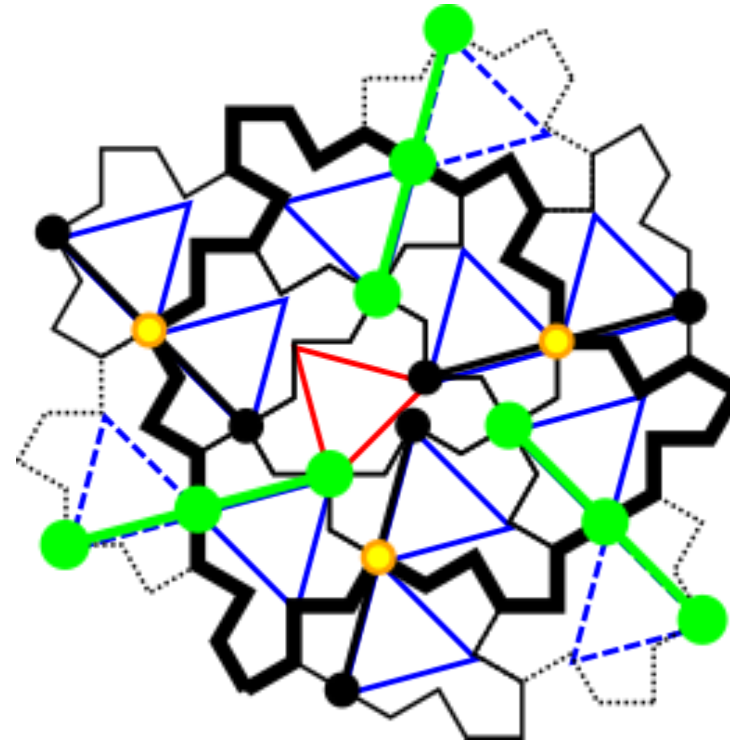
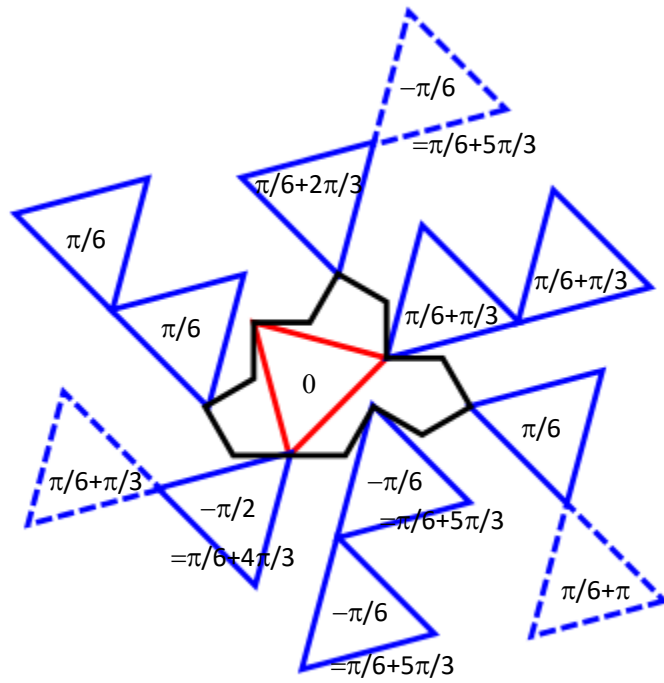
Triangles inside a full cluster



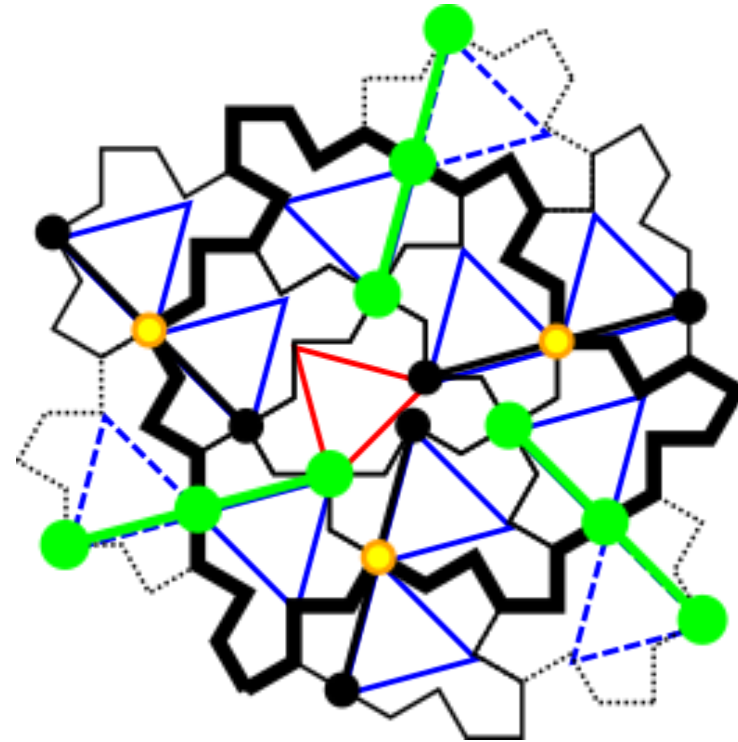
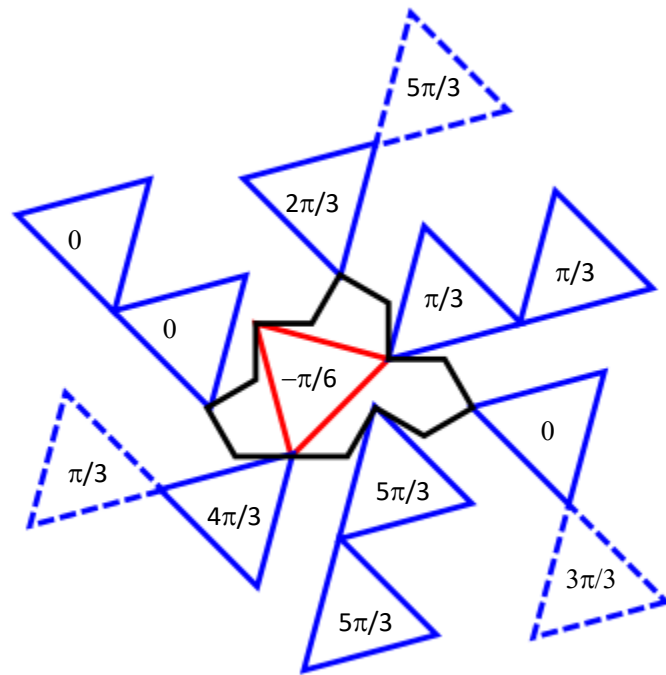
Identification of junction lines



Sequence of rotations

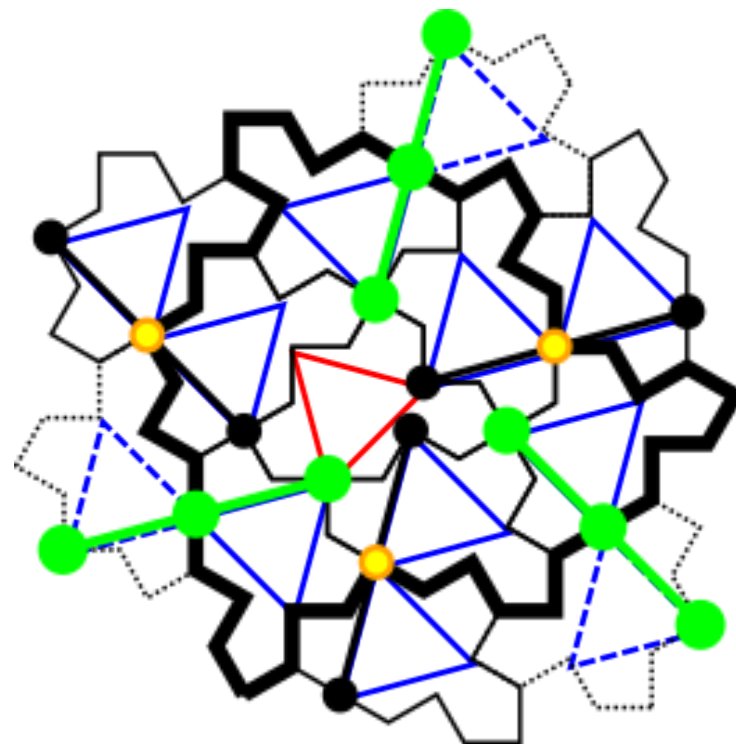
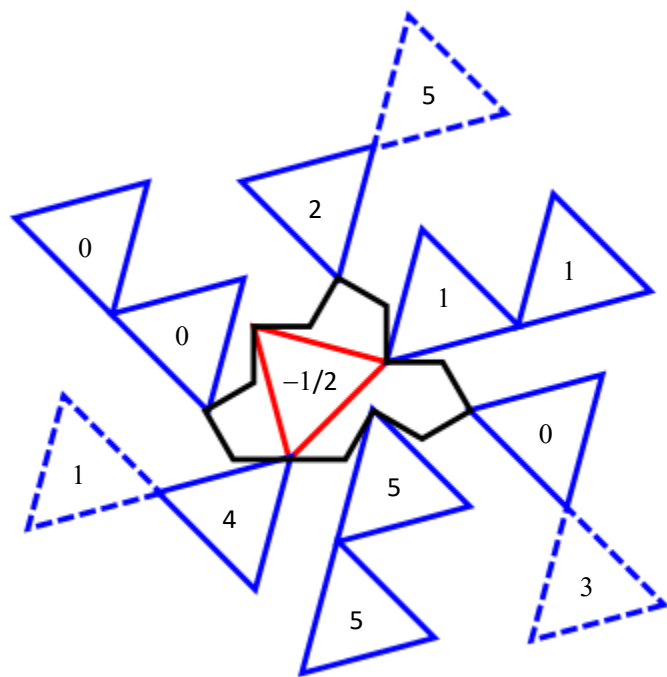


Sequence of rotations



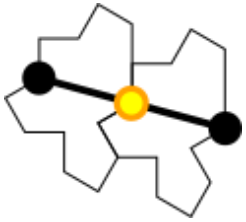
Sequence of rotations

$\pi/3=2\pi/6$
factorization

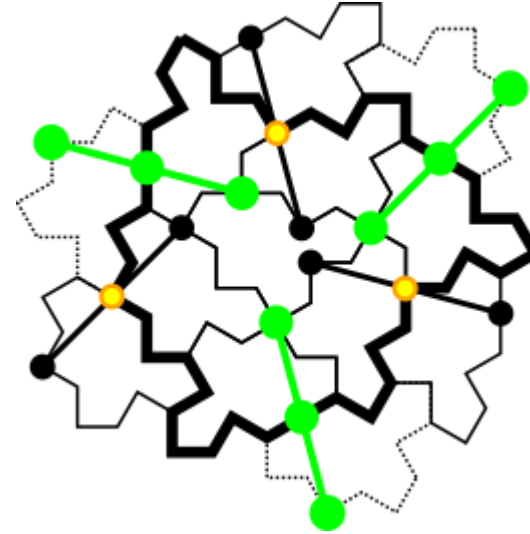
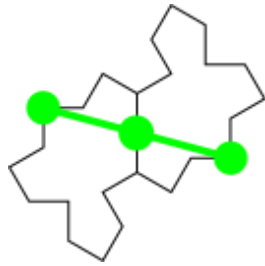


Junction lines between full clusters

translation

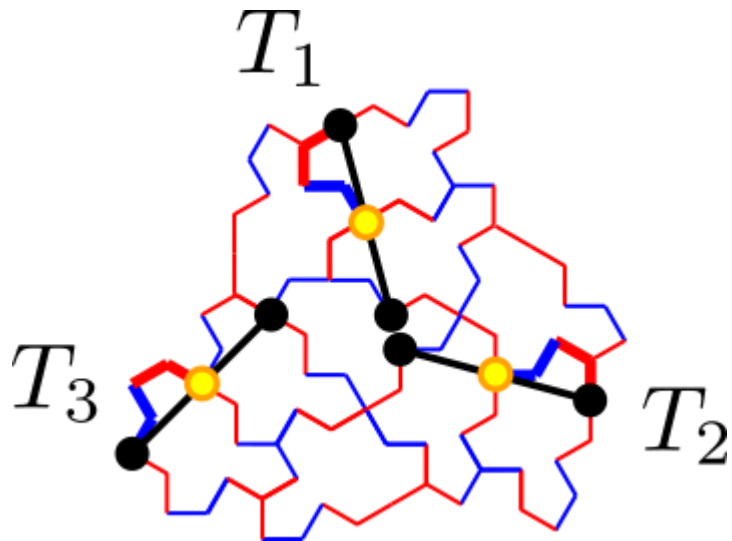


inversion



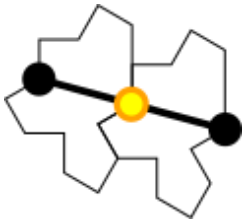
Three black points and three green points on the odd monotile in the middle.
They are located always at the same positions.

Three even monotiles outside the C cluster in dashed lines obtained by inversion.

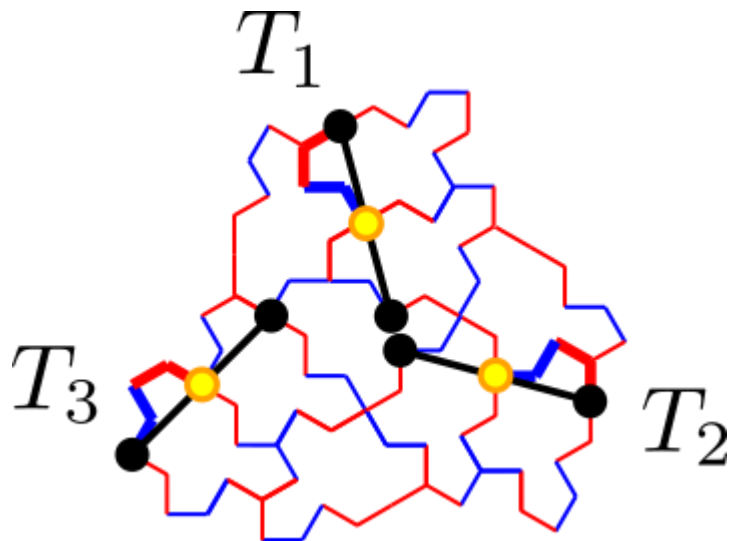
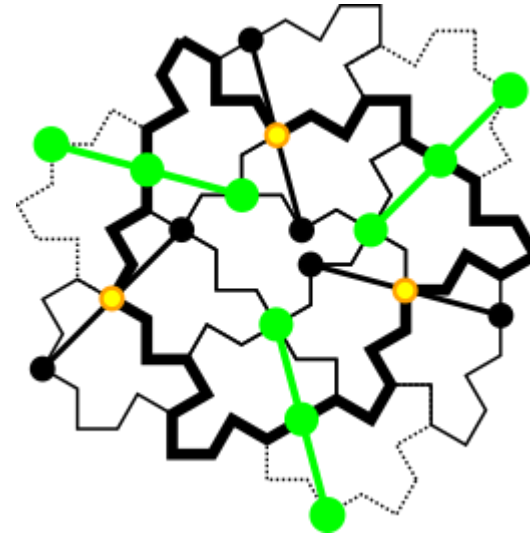
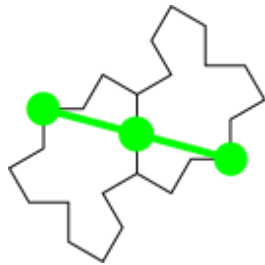


Junction lines

translation



inversion



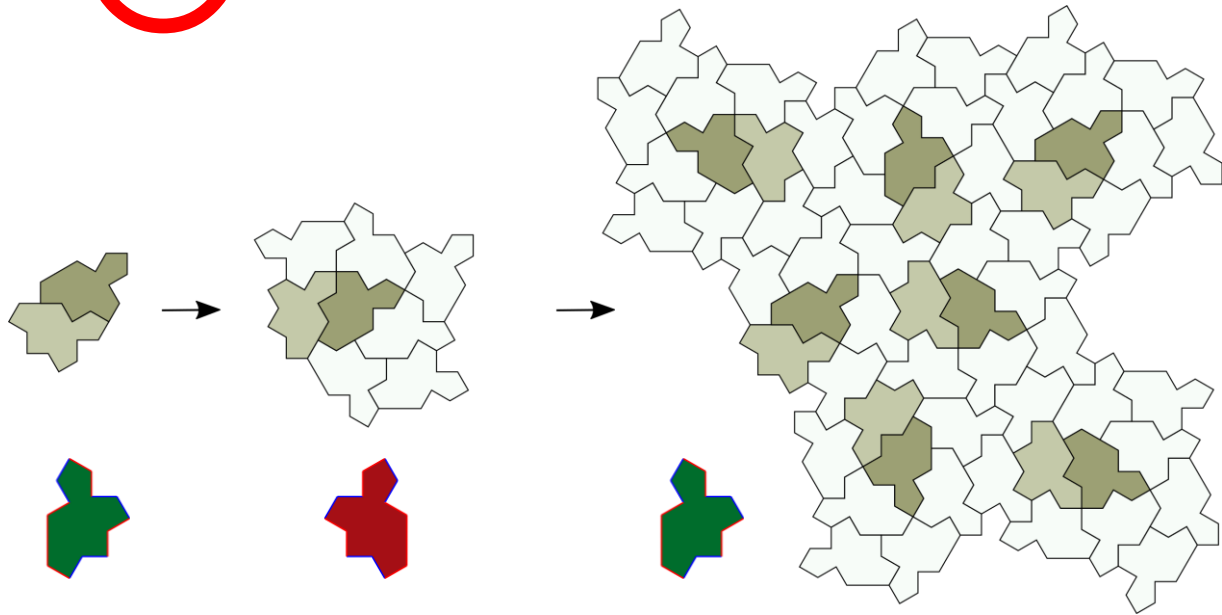
Black junction lines == translation == always present

when two clusters are assembled, some green lines may be superimposed on black ones !!!
-> black one is the good one
-> translation and not inversion

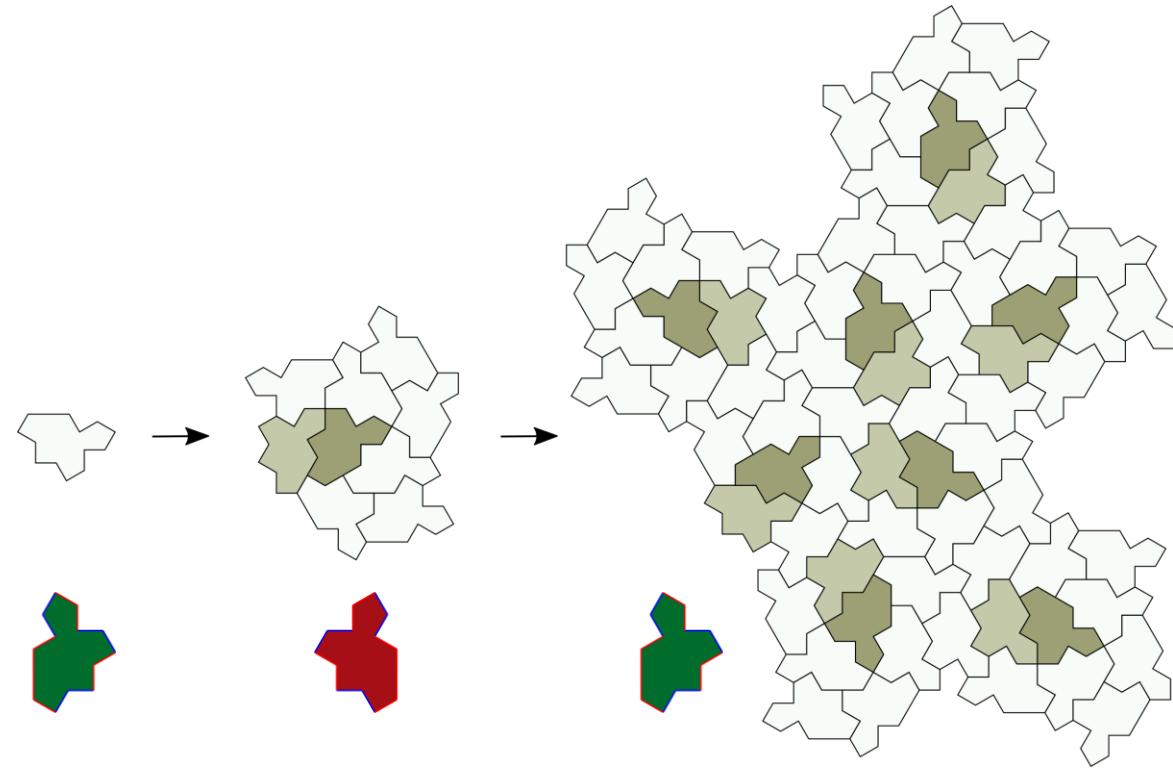
Inflation with alternate chirality of Tile(1,1) at each step



<https://cs.uwaterloo.ca/~csk/spectre/app.html>



-> Need for homochiral inflation
to get the orientation distribution
at each inflation step

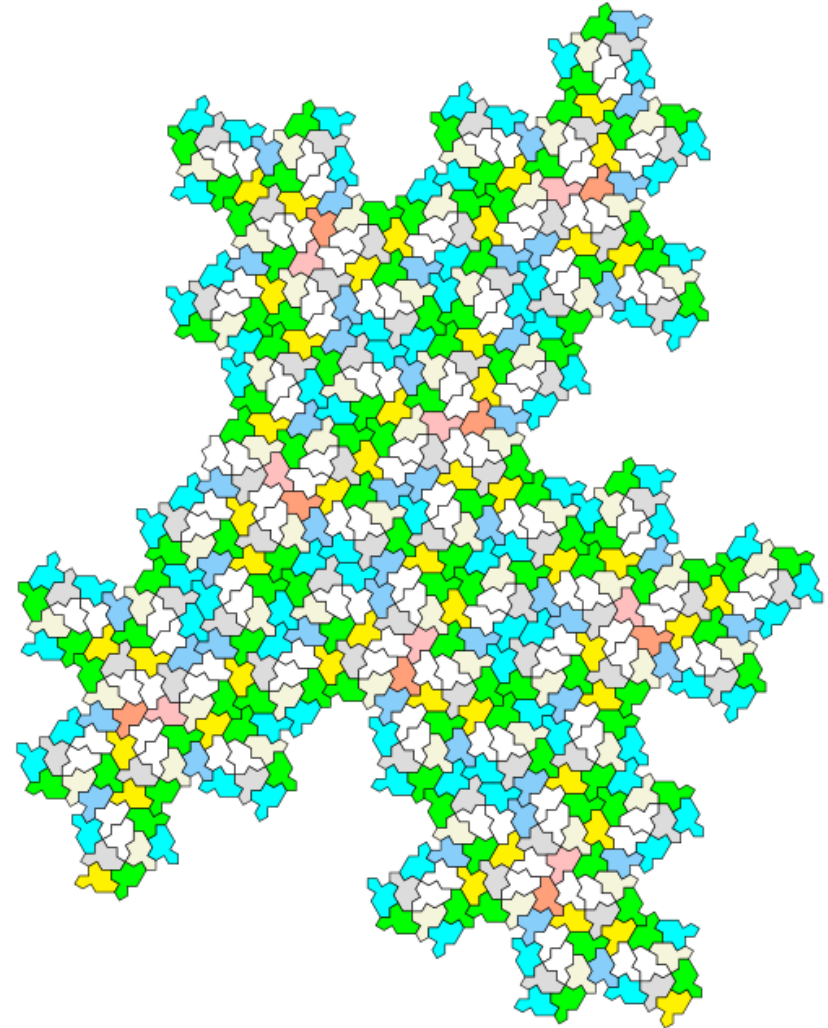


Python script to generate the tiling

Drawsvg python library

<https://pypi.org/project/drawsvg/>

<https://github.com/shrx/spectre/blob/master/spectre.py>



Python script: step 1

svg from python code

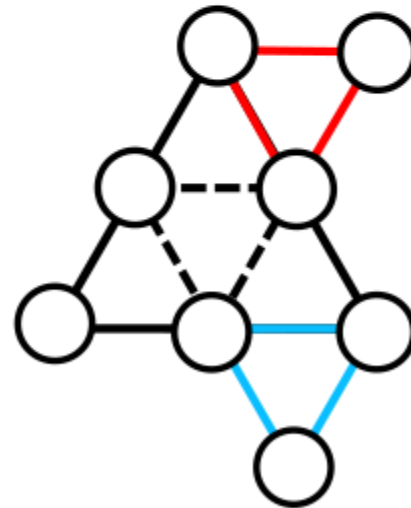
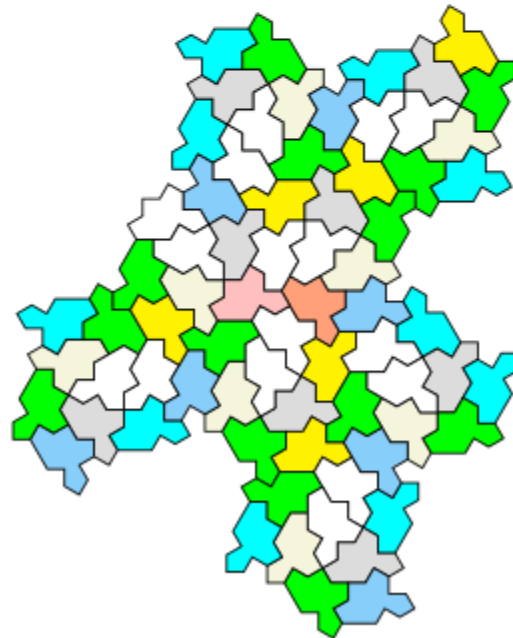
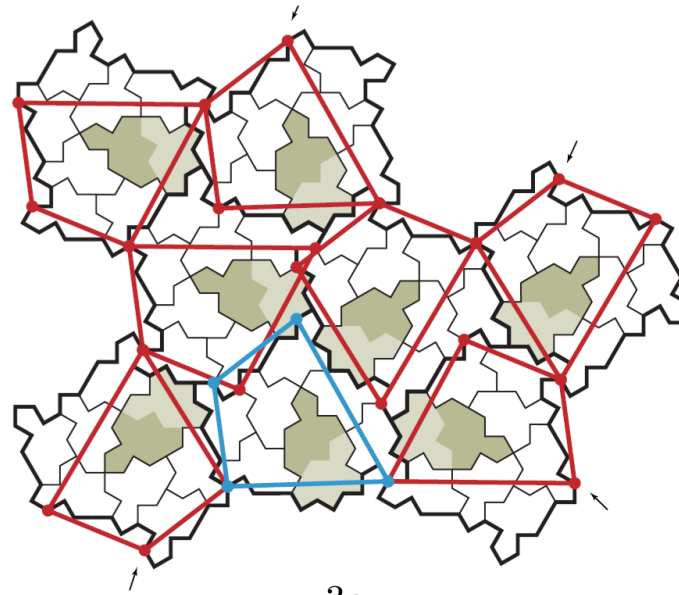
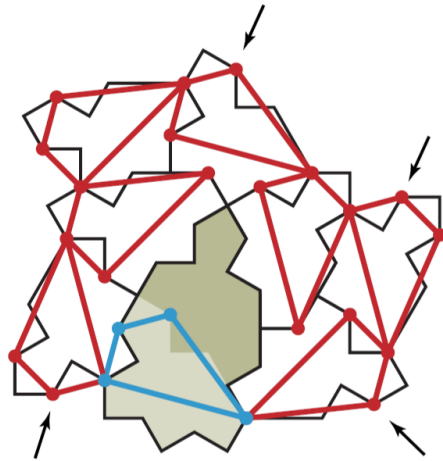
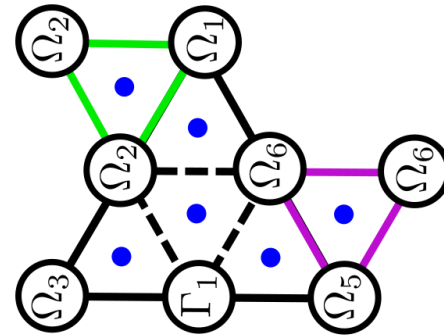
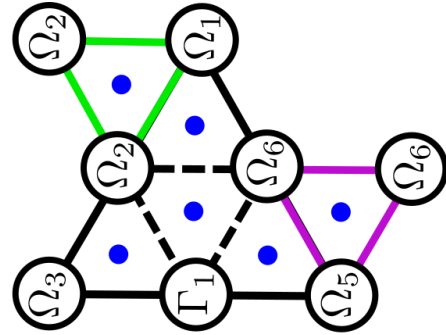
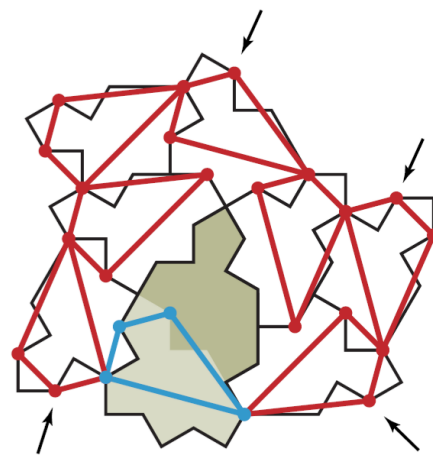
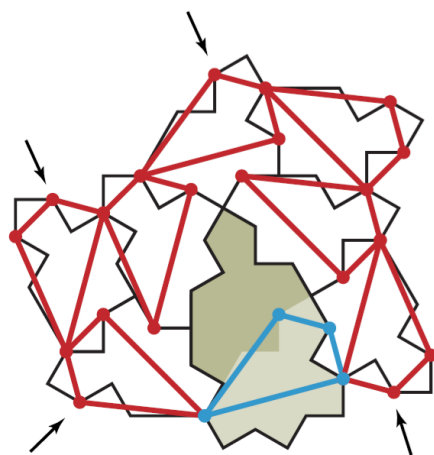
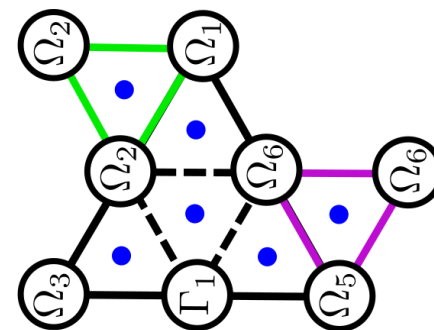
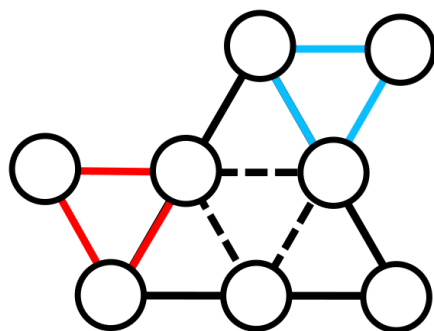


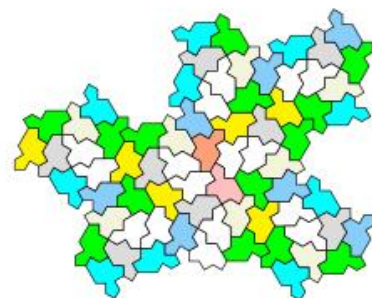
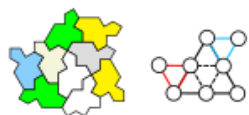
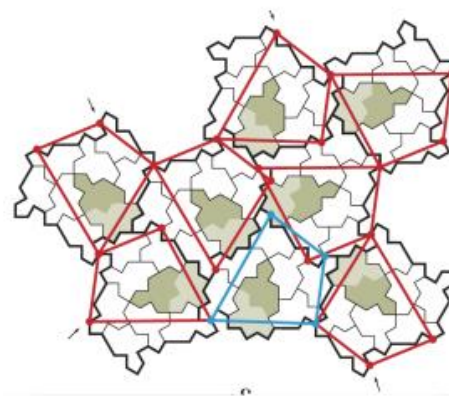
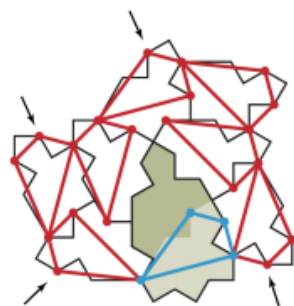
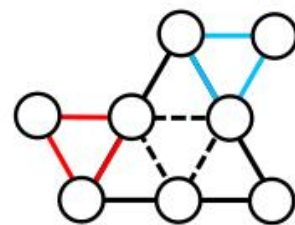
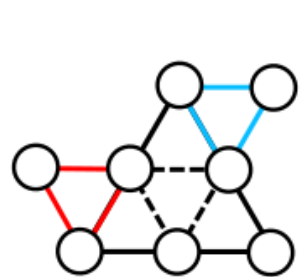
Fig A.1 (annexe)



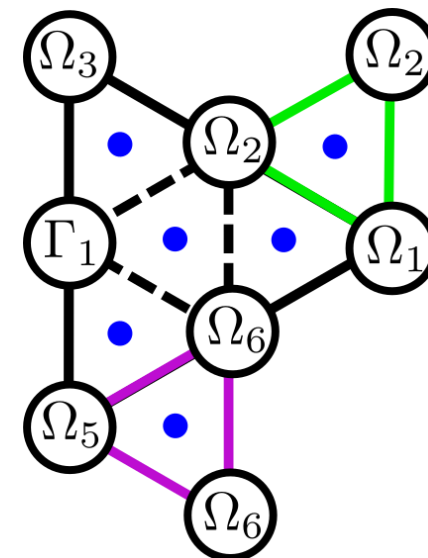
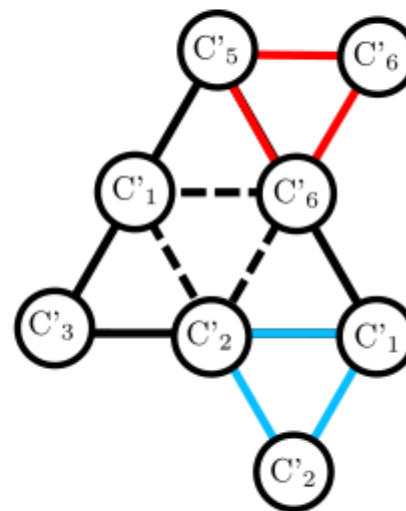
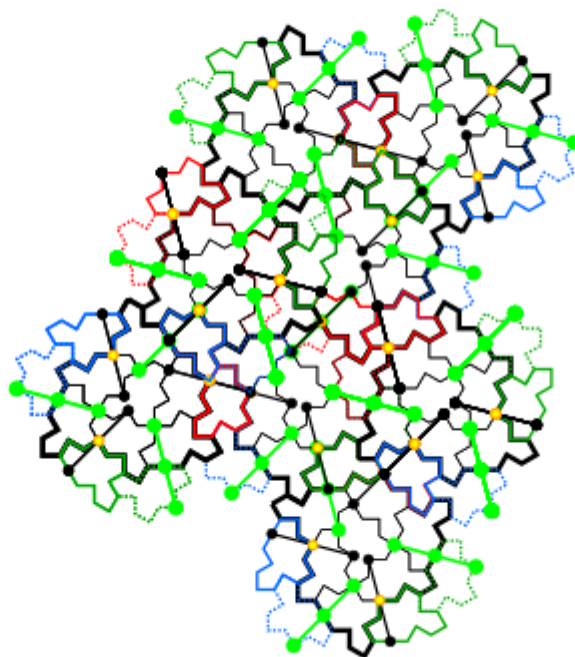
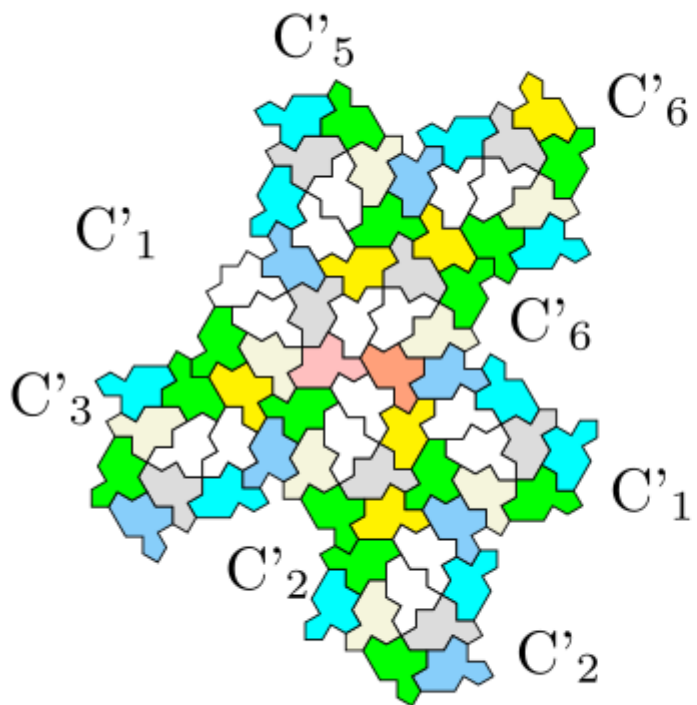
Comparison with Fig A.1 (annexe)



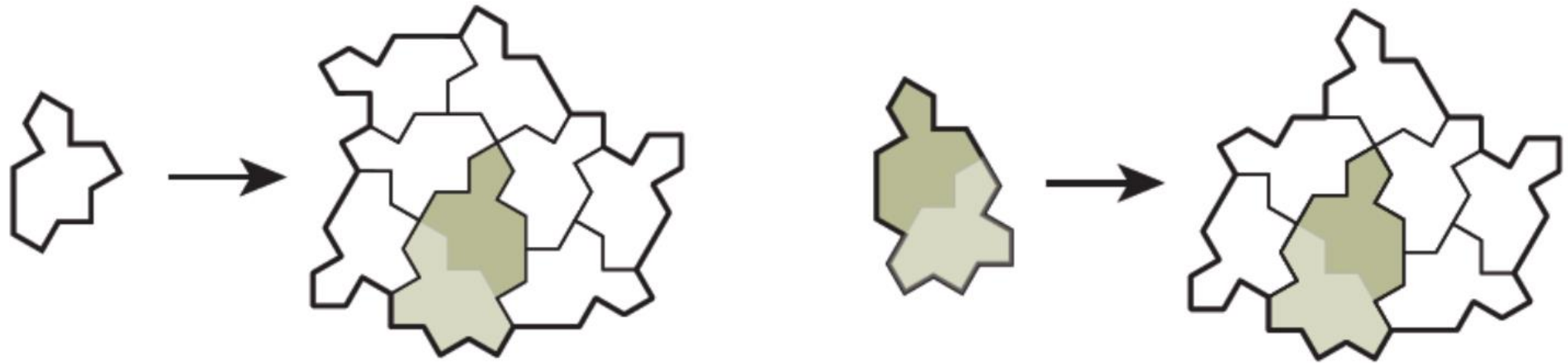
Comparision with Fig A.1 (annexe)



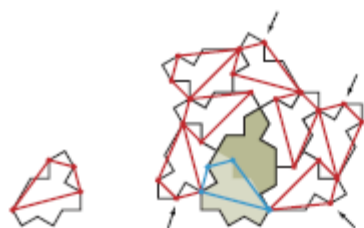
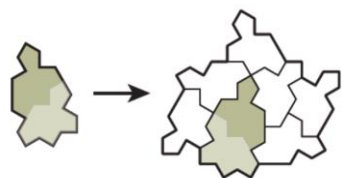
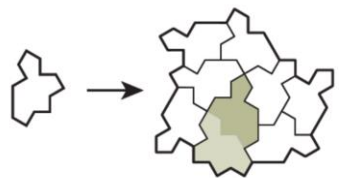
Sequence of eight clusters



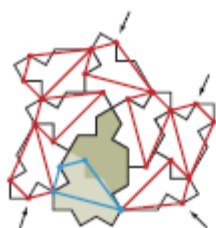
Substitution rule



Key points



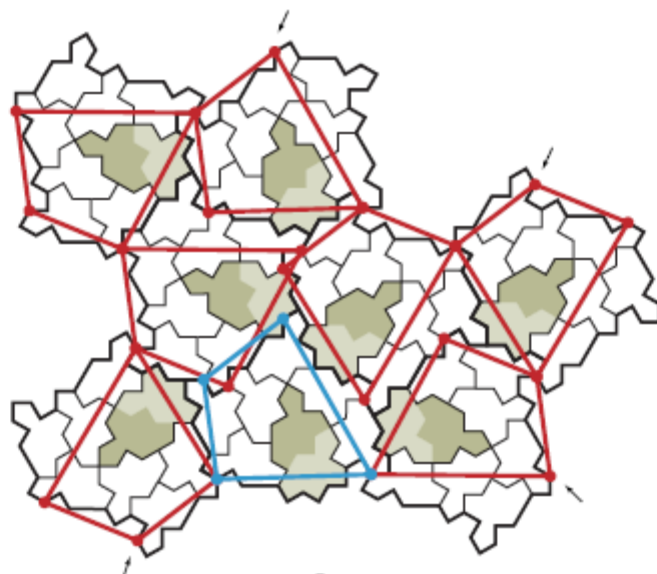
1.



2a.



2b.



3a.



3b.